Lesson 1 — Extra Practice

The Cartesian Plane, Growing Patterns, and Types of Variables

Part A — Growing Patterns Whose Rule Is Not Obvious

1. In each figure below a solid square block of *grey* tiles is surrounded by a single ring of *white* tiles. Let n be the figure number and let W be the number of **white** tiles. Notice that W does not go up by one at each step.

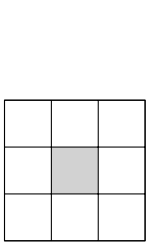


Figure 1

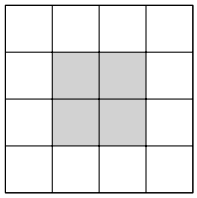


Figure 2

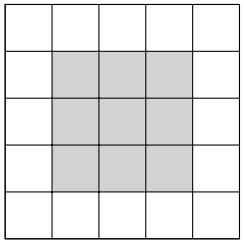


Figure 3

(a) Complete the table.

Figure Number, n	1	2	3	4	5
White Tiles, W					

(b) By how much does W increase from one figure to the next? Point to the four tiles on the diagram that account for it.

(c) Write an equation for W in terms of n.

(d) Figure n is an $(n+2)$ by $(n+2)$ square with an n by n grey centre. Use that to get the same equation a second way.

(e) How many white tiles are in Figure 25?

(f) Which figure uses exactly 96 white tiles?

(g) Explain why *no* figure in this pattern uses exactly 90 white tiles.

2. The tile pattern below **starts at Figure 0**. Let n be the figure number and T the total number of tiles.



Figure 0

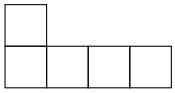


Figure 1



Figure 2

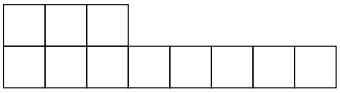


Figure 3

(a) Complete the table. Note where the figure numbers begin.

Figure Number, n	0	1	2	3	4	5
Total Tiles, T						

(b) Describe the pattern in words by describing the top row and the bottom row of Figure n separately.

(c) Write an equation for T in terms of n.

(d) A student who assumed the pattern started at Figure 1 wrote $T = 3n - 1$. Test both equations at $n = 0$ and at $n = 3$ and say which is right.

(e) How many tiles are in Figure 40?

(f) Which figure has exactly 59 tiles?

(g) In your equation, what does the constant term count, and where is it in every diagram?

3. Toothpicks are laid out to make a chain of hexagons. Each new hexagon shares one complete side with the hexagon before it. Let n be the number of hexagons and H the number of toothpicks.

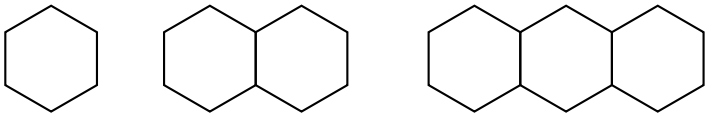


Figure 1 Figure 2 Figure 3

(a) Count the toothpicks in each figure and complete the table.

Hexagons, n	1	2	3	4	5
Toothpicks, H					

(b) The first hexagon needs 6 toothpicks but each one after that needs fewer. How many fewer, and why?

(c) Write an equation for H in terms of n.

(d) How many toothpicks does a chain of 32 hexagons need?

Parts (e) and (f) run the rule backwards: you are given the count and must find the figure number.

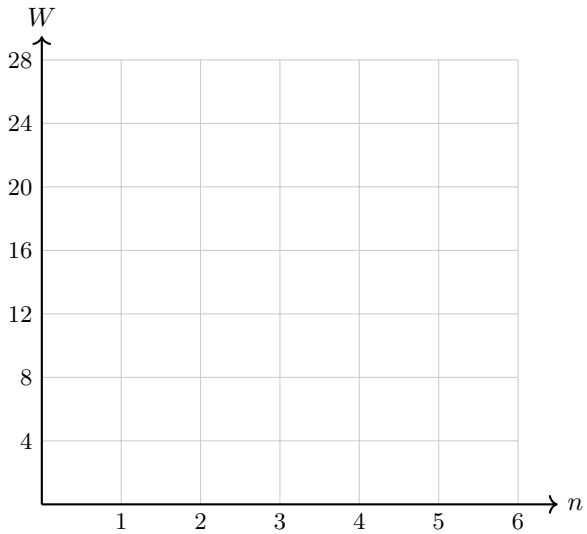
(e) A chain uses 246 toothpicks. How many hexagons does it have? Show the algebra.

(f) Could a chain use exactly 300 toothpicks? Solve, then explain what your answer means.

(g) Every possible value of H leaves the same remainder when divided by 5. What is it, and why?

Part B — Plot It, Then Read It Back

4. Plot the data from your table in question 1 on the grid below.



(a) State the coordinates of the point that represents Figure 5.

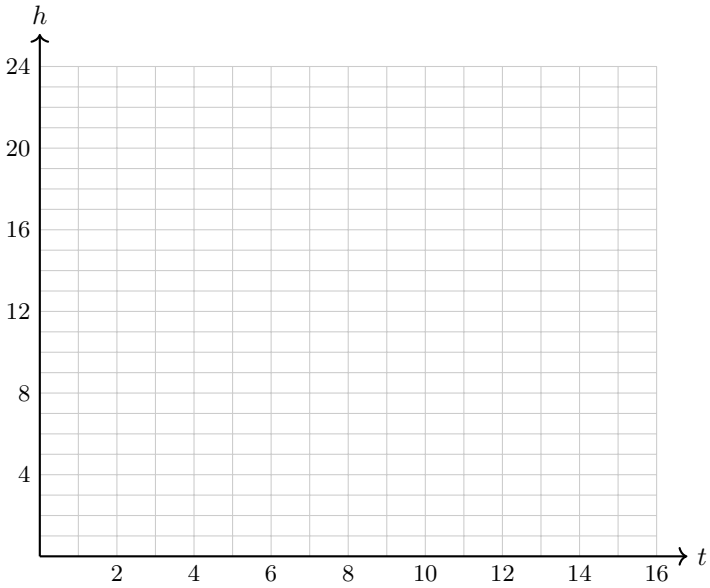
(b) Should these points be joined? Answer using the rule stated in question 7.

(c) The plotted points all lie in one quadrant. Which one, and why can no point of this pattern lie in any other?

5. A candle is 24 cm tall when it is lit and burns down at a steady 1.5 cm every hour. Let t be the time in hours and h the height of the candle in centimetres.
(a) Complete the table.

Time, t **(h)**	0	4	8	12	16
Height, h **(cm)**	24				

(b) Plot the five points and join them.



(c) **Read off** the height of the candle at $t = 6$ h.

(d) **Read off** the time at which the height is 9 cm.

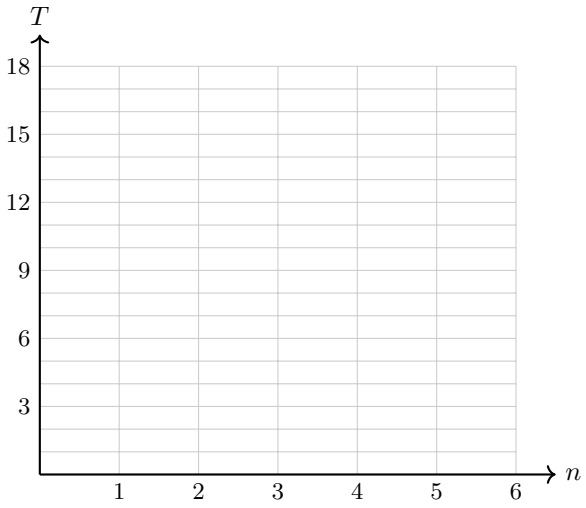
(e) Write an equation for h in terms of t.

(f) Use the equation to calculate the two values you read off in (c) and (d). Do they agree with your readings?

(g) The graph should stop somewhere. At what value of t, and why does the relation stop making sense past it?

(h) Is a reading taken off a graph ever better than a calculated value? Say what each one is good for.

6. Plot the data from your table in question 2 on the grid below — this time the horizontal axis must start at $n = 0$.



(a) Read the number of tiles in Figure 4 off your plot, then calculate it from your equation. Do the two agree?

(b) One plotted point sits on the vertical axis. Give its coordinates and say what it means in the tile pattern.

(c) Your six points lie in a straight row, yet you must not draw the line. Explain.

Part C — Discrete or Continuous? It Depends What You Measure

7. The booklet settles this with one rule:

> *"Connect the points when the variables are continuous, i.e. every in-between value has meaning; leave them unconnected when the variables are discrete."*

Classify each variable below as **discrete** or **continuous**, and in each case name one in-between value and say whether it has meaning.

(a) the number of passengers on a bus

(b) the distance the bus travels on its route

(c) the volume of fuel in the bus's tank

(d) the number of stops the bus makes

(e) the time the trip takes

(f) the number of seats still empty

8. Each pair below is about the *same* underlying quantity, but the two versions are measured differently. Classify each version, and in one sentence say what makes the pair split.

(a) **Flour.**
i) the mass of flour scooped from an open bin
ii) the number of sealed 2 kg bags of flour bought

(b) **A phone call.**
i) how long the call lasts, timed by a stopwatch
ii) the number of whole minutes the call is billed for

(c) **Oven heat.**
i) the actual temperature inside the oven as it warms up
ii) the setting on a dial that clicks only to 25° marks

(d) **Water.**
i) the amount of water in a jug
ii) the number of identical cups the jug completely fills

9. The same table of values is used for two different situations.

x	1	2	3	4	5
y	12	24	36	48	60

(a) x is the number of identical crates loaded onto a truck and y is their total mass in kilograms. Join or not? Why?

(b) x is the number of hours a worker has been on shift and y is the pay earned, paid by the minute at \$12 an hour. Join or not? Why?

(c) The numbers are identical and the equation is $y = 12x$ both times. Say precisely what decides the answer, and why it is not the numbers.

Part D — Quadrants From Sign Conditions

10. A point $P(x, y)$ satisfies the condition given. Name the quadrant it must lie in. If more than one quadrant is possible, name them all; if P lies on an axis, say which axis and state that it is in **no** quadrant.

(a) $x > 0$ and $y < 0$ (b) $xy > 0$ and $x < 0$ (c) $xy < 0$ and $y > 0$

(d) $x = 0$ and $y < 0$ (e) $\dfrac{x}{y} > 0$ (f) $x + y = 0$ and $x > 0$

11. The point $P(a, b)$ lies in Quadrant III. State the quadrant of each point below and justify each with the signs of a and b.

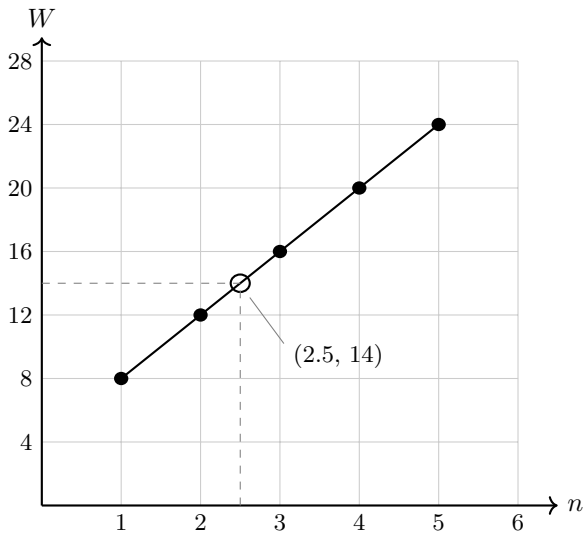
(a) (b, a) (b) $(-a, b)$ (c) $(a, -b)$ (d) $(-a, -b)$

12. For each point, state its quadrant, or name the axis it lies on and state that it belongs to no quadrant.

(a) $A(4, 0)$ (b) $B(-3, -7)$ (c) $C(0, -5)$ (d) $D(6, -2)$ (e) $E(-1, 8)$ (f) $F(0, 0)$

Part E — Error Analysis

13. *(Error analysis)* A student plotted the white-tile pattern from question 1 and then joined the points with a ruler, as shown.



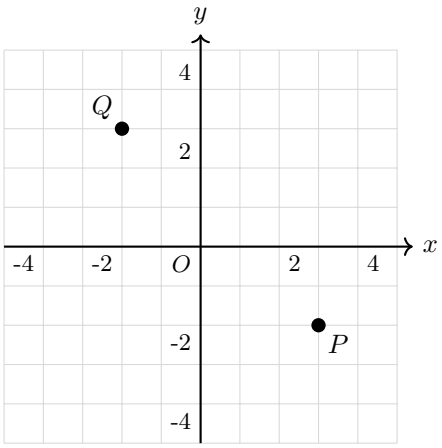
(a) The hollow point sits on the student's line. Write down what it claims, in words about tiles and figures.

(b) Give **two** separate reasons that claim is impossible — one about n, one about W.

(c) Solve $4n + 4 = 14$ and use the result to back up your answer to (b).

(d) Describe exactly what the finished graph should look like instead, and state the booklet's rule that says so.

14. *(Error analysis)* Asked to write down the coordinates of the marked point P on the grid below, a student wrote $P(-2, 3)$. The point they actually described is marked Q.



(a) State the correct coordinates of P, and describe the walk from the origin that reaches it.

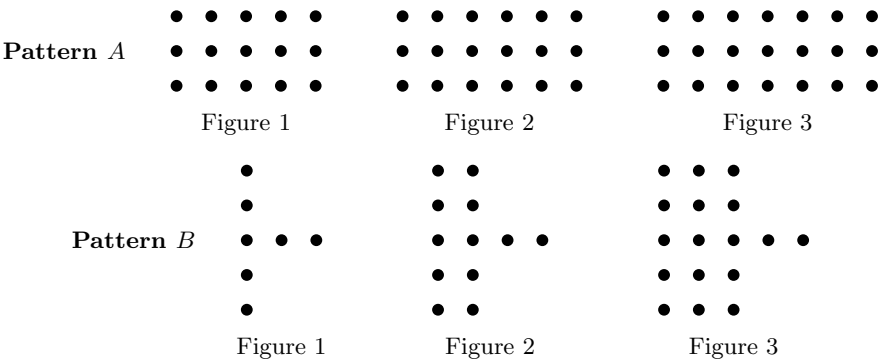
(b) Name the quadrant of P and the quadrant of Q.

(c) The student swapped the numbers *and* the signs travelled with them. Write one sentence of advice that would stop this mistake for good.

(d) For which points does swapping the two coordinates make no difference at all? Describe them, and give two examples.

Part F — A Stretch, and Two for the Test

15. *(Stretch)* Two dot patterns grow side by side. Pattern A starts larger, but Pattern B grows faster.



(a) Count the dots in each figure and complete both rows of the table.

Figure Number, n	1	2	3	4	5	6
Pattern A dots						
Pattern B dots						

(b) Pattern A is always three rows of dots. Use that to write an equation for A in terms of n.

(c) Pattern B is n full columns plus a fixed tail. Write an equation for B in terms of n.

(d) At which figure number do the two patterns have the same number of dots? Solve $A = B$ and show every step.

(e) How many dots does each pattern have at that figure? Check the answer in both equations.

(f) For which figure numbers does A have more dots than B? Use your table to say how you know.

(g) The solution of $A = B$ happened to be a whole number. Explain why a non-whole-number solution would have meant the patterns never actually match.

16. *(Multiple Choice)* A point $P(x, y)$ satisfies both $xy < 0$ and $y > x$. The point P must lie in
A. Quadrant I B. Quadrant II C. Quadrant III D. Quadrant IV

17. *(Numerical Response)* In a growing pattern, Figure 1 contains 9 dots and every figure after it contains 6 more dots than the figure before.

The number of the figure that contains exactly 237 dots is _____.
(Record your answer in the numerical response box from left to right.)

Lesson 2 — Extra Practice

Describing Relationships Between Two Quantities

Part A — One Relation, Four Descriptions

1. **Start from words.** A rain barrel holds 200 L of water. A soaker hose drains it at a steady rate of 25 L per minute.

(a) Name the independent variable and the dependent variable, and state the rule you used to decide.

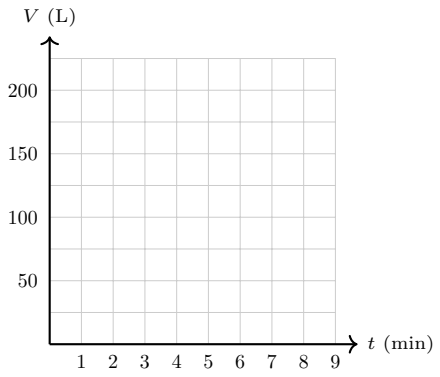
(b) Complete the table of values. Use t for the time in minutes and V for the volume in litres.

t	0	1	2	3	4
V					

(c) Write the relation as a set of ordered pairs.

(d) The water level falls continuously, not in jumps. What does that tell you to do with the points after you plot them?

(e) Plot the ordered pairs on the grid and complete the graph.



(f) Use your graph to find the time at which the barrel is empty.

(g) Write an equation for V in terms of t, and use it to check your answer to (f).

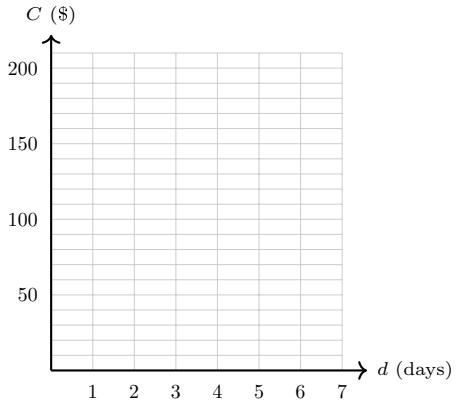
2. **Start from a table.** A tool shop rents out a pressure washer. The table shows the total charge C, in dollars, for d days of rental. **Notice that the inputs do not go up by 1 each time** — you cannot just read the pattern off the bottom row.

d	1	2	4	6
C	50	80	140	200

(a) Describe this relation in words: say what the shop charges up front and what it charges per day. Explain how the table told you each number.

(b) Write the relation as a set of ordered pairs, and state which variable each coordinate holds.

(c) Plot the four ordered pairs on the grid.



(d) The shop rents only in whole days. Should you join your points with a line? Explain.

(e) Determine the charge for 3 days and for 10 days.

(f) A customer is charged \$440. How many days did they keep the washer?

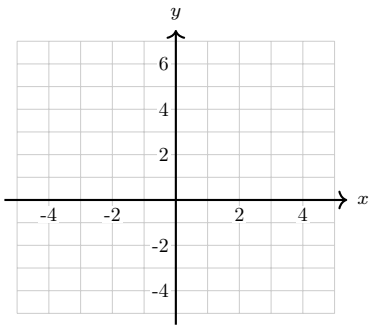
3. **Start from ordered pairs.** A relation is given by $(-2,-3),\ (-1,-1),\ (0,1),\ (1,3),\ (2,5)$.

(a) Build a table of values from the pairs.

x					
y					

(b) Draw a mapping diagram for this relation. Label the ovals X and Y.

(c) Graph the relation.

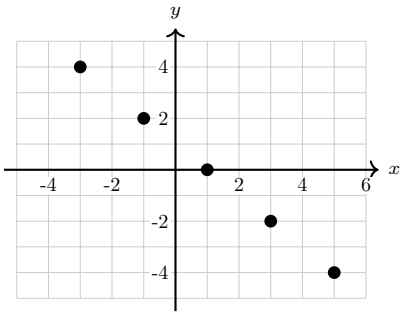


(d) Describe the relation in words, beginning "each output is . . . "

(e) Write an equation for the relation.

(f) Determine the output when the input is 6, and the input that gives an output of -9.

4. **Start from a graph.** The graph below shows every ordered pair in a relation.



(a) Read the five ordered pairs off the graph.

(b) Write the relation as a table of values.

x					
y					

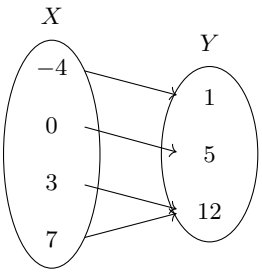
(c) Draw a mapping diagram for the relation.

(d) Describe the relation in words, then write an equation for it.

(e) The points were left unjoined. What does that tell you about the relation, and what would joining them have claimed?

Part B — Mapping Diagrams, Both Directions

5. Use the mapping diagram to answer the questions.



(a) List the inputs, then list the outputs.

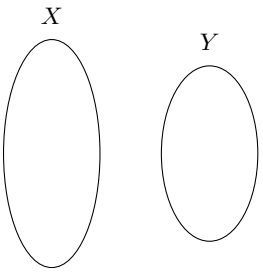
(b) Write the relation as a set of ordered pairs.

(c) One element of Y has two arrows pointing at it. Which one, and what does that look like in your list of ordered pairs?

(d) Is there an element of X with two arrows leaving it? Explain why that difference matters when you write the pairs.

6. A relation is given by the ordered pairs $(3, 12),\ (5, 20),\ (7, 12),\ (9, 28)$.

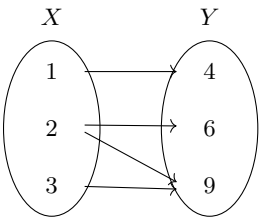
(a) Fill in the ovals and draw the arrows. Write each output **once only**, in increasing order.



(b) How many arrows did you draw, and how many numbers did you write in the oval on the right? Why are those two counts different?

(c) Name the independent variable and the dependent variable, and say how the diagram shows which is which.

7. Not every relation gives one output per input. Use the mapping diagram below.



(a) Write the relation as a set of ordered pairs.

(b) Which input has more than one output? Explain why this relation still counts as a relation even though you cannot write "the output for that input."

(c) Reverse every arrow, so each old output becomes an input. Write the ordered pairs of this new relation, then draw its mapping diagram.

Part C — Tables With No Constant Difference

8. A square patio is built with side length s metres. Its area is A square metres.

(a) Complete the table, then fill in the row of **first differences** (each entry is the area minus the previous area).

s	1	2	3	4	5	6
A						
first difference	—					

(b) In a linear relation the first differences are all equal. Are yours? What does your answer prove about this relation?

(c) Describe the relation in words, then write an equation for A in terms of s.

(d) The side length doubles from 3 m to 6 m. Does the area double? State what it does instead, using your table.

(e) Determine the area of a patio with side length 12 m.

(f) A patio has area 196 m^2. Determine its side length.

9. A relation is given by the table below.

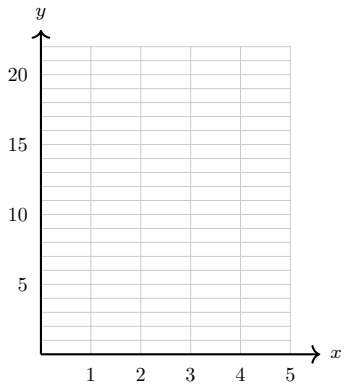
x	0	1	2	3	4
y	5	6	9	14	21

(a) Write out the first differences. Is the relation linear? Justify your answer.

(b) Each output is a fixed number more than the square of the input. Write the equation.

(c) Determine y when $x = 7$.

(d) Plot the five ordered pairs from the table and join them with a smooth curve.



(e) Determine an input that produces an output of 41.

(f) There is a **second** input that also gives 41. Find it, and explain why a squaring rule does this but a rule like $y = 3x + 2$ never does.

10. Two relations are given below. Exactly one of them is linear.

Relation P

x	1	2	3	4
y	7	12	17	22

Relation Q

x	1	2	3	4
y	3	12	27	48

(a) Write the first differences for each relation and say which relation is linear.

(b) Write an equation for each relation. One involves a square.

(c) Determine the output of each relation when the input is 10.

Part D — Which Variable Depends on Which

11. For each pair of quantities, name the independent variable and the dependent variable, then complete the sentence "the _____ depends on the _____."

(a) the number of litres of fuel pumped, and the cost of the fill-up

(b) the diameter of a round pizza, and its area

(c) the number of weeks since a sunflower was planted, and its height

(d) a speed skater's average speed, and the time they take to finish 500 m

(e) **Two** of (a) to (d) are connected by a formula you already know. Name one and give its formula.

12. Naming the independent variable is a decision about the *situation*, not about the letters. For each pair below, state which quantity you would put on the horizontal axis and why.

(a) tickets sold at a school play, and the total money collected

(b) a child's age in months, and the child's mass

(c) the number of lifeguards on duty, and the number of swimmers in the pool

(d) the distance a taxi drives, and the fare charged

(e) One of the four can be argued in *both* directions. Identify it and give the argument each way.

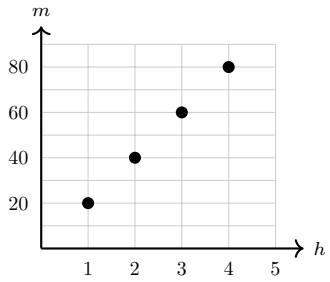
Part E — Find the Mistake

13. *(Error analysis)* A student is told: "a summer job pays \$20 per hour; relate the number of hours worked, h, to the money earned, m." Their table is correct:

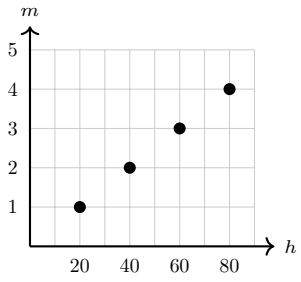
h	1	2	3	4
m	20	40	60	80

But they write the ordered pairs as $(20, 1)$, $(40, 2)$, $(60, 3)$, $(80, 4)$ and produce Graph 2 below.

Graph 1



Graph 2

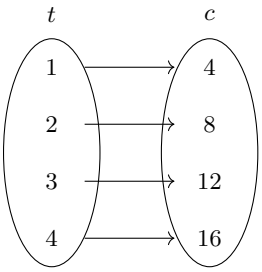


(a) State the rule about ordered pairs that the student broke, and write the four pairs correctly.

(b) Read Graph 2 literally, using the axis labels printed on it. What does the point closest to the m-axis claim about this job?

(c) Which graph is correct, and what single change to Graph 2 would fix it?

14. *(Error analysis)* A banquet hall seats 4 people at every table. The mapping diagram below relates the number of tables, t, to the number of chairs, c. A student says: "the number of tables depends on the number of chairs, so the ordered pairs are $(4,1),\ (8,2),\ (12,3),\ (16,4)$."



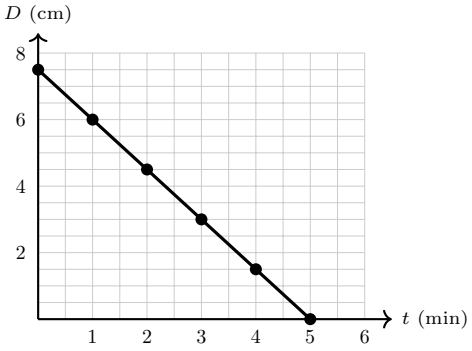
(a) State the rule about arrows in a mapping diagram that the student ignored.

(b) Write the correct ordered pairs, and write an equation for c in terms of t.

(c) Draw the mapping diagram that *would* match the student's ordered pairs, and write the equation it represents.

Part F — A Stretch, and Two for the Test

15. The graph shows the depth of water, D (cm), in a leaking bucket t minutes after the leak started. **The grid lines are 0.5 units apart**, so some readable values are not whole numbers.



(a) Describe this relation in words. Say how deep the water was when the leak started and how fast the depth is falling.

(b) Name the independent variable and the dependent variable.

(c) The graph passes exactly through a grid intersection at six points. Record them in the table.

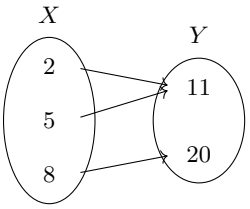
t						
D						

(d) Write the six ordered pairs, and write an equation for D in terms of t.

(e) The graph stops at $t = 5$ and is not drawn further. Give the reason from the situation, not from the equation.

(f) Use your equation to find D when $t = 2.5$, and explain why you could not have read that value off the grid.

16. *(Multiple Choice)* Which set of ordered pairs represents the relation shown in the mapping diagram?



A. $(2, 11),\ (5, 11),\ (8, 20)$
B. $(11, 2),\ (11, 5),\ (20, 8)$
C. $(2, 11),\ (5, 20),\ (8, 20)$
D. $(2, 20),\ (5, 11),\ (8, 11)$

17. *(Numerical Response)* A relation is described in words: "the output is 4 more than the square of the input." Let A be the output when the input is 7, and let B be the positive input that produces an output of 104. The value of $A + B$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 3 — Extra Practice

Where Graphs Meet the Axes: x- and y-Intercepts

Part A — Read the Question: a Value or an Ordered Pair?

*Lesson 3 asks for intercepts two different ways. "Determine the **value** of the y-intercept" wants a single number, such as -8. "Answer as **ordered pairs**" wants a point, such as $(0, -8)$. Every question below says which one it wants. Give exactly that.*

1. Determine the **value** of the x-intercept and the **value** of the y-intercept of each line.

(a) $5x - 2y = 30$

(b) $\frac{x}{4} + \frac{y}{3} = 1$

(c) $2x - 7y - 21 = 0$

(d) $y = \frac{2}{3}x - 6$

(e) $0.25x + 0.4y = 2$

(f) $3(x - 2) = 4(y + 1)$

2. Determine the x-intercept(s) and the y-intercept of each parabola. **Answer as ordered pairs.** If a graph has no x-intercept, write *none*.

(a) $y = x^2 - 5x - 14$

(b) $y = 2x^2 - 18$

(c) $y = x^2 - 6x + 9$

(d) $y = x^2 + 4x + 7$

(e) $y = -x^2 + 2x + 8$

(f) $y = 3x^2 - 12x$

3. A line has equation $3x + 5y = 60$.

(a) State the **value** of the x-intercept and the **value** of the y-intercept.

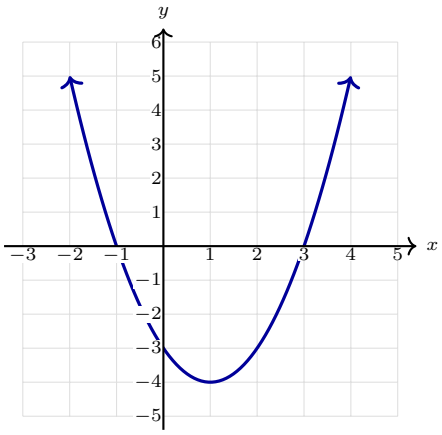
(b) State the same two intercepts **as ordered pairs**.

(c) Only your answers in (b), not those in (a), could be marked as dots on the grid. Explain why, and say what extra information the ordered pair carries.

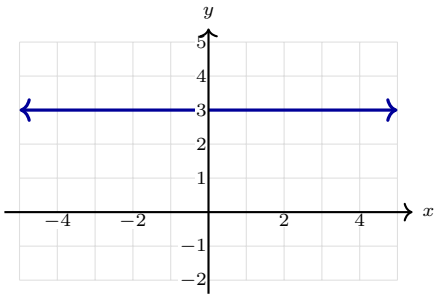
4. Each statement below is careless. Say what is wrong with it, then rewrite it so a reader knows *which* axis and *which* form is meant.
(a) "The intercept is 5." (b) "The graph crosses at 3." (c) "x-intercept $= (0, 7)$."

Part B — Reading Intercepts Off a Graph

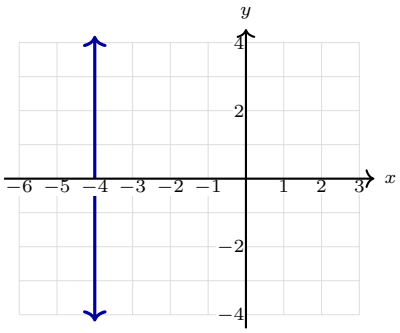
5. State the x-intercept(s) and the y-intercept(s) of each graph **as ordered pairs**. If a graph has no intercept of one kind, write *none* and say in a few words why.
(a)



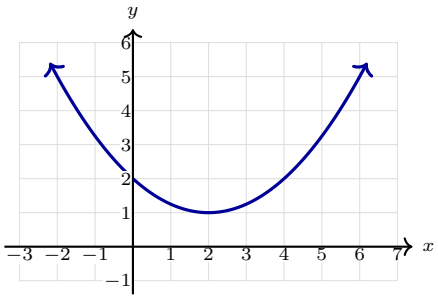
(b)



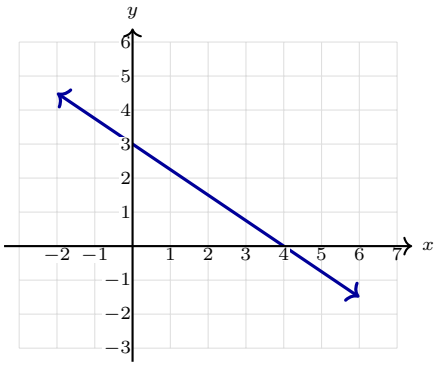
(c)



(d)



6. The line below is the graph of $3x + 4y = 12$.



(a) Read the **value** of the x-intercept and the **value** of the y-intercept off the graph.

(b) Confirm both by substituting into $3x + 4y = 12$.

(c) State the two intercepts **as ordered pairs**.

(d) A classmate answers (a) with "$(4, 0)$ and $(0, 3)$." Their points are right. Explain why it is still not an answer to the question asked.

Part C — When an Intercept Does Not Exist

7. Determine the x-intercept(s) and the y-intercept(s) of each graph **as ordered pairs**. Write *none* where there is no such intercept, and describe the graph in a few words.

(a) $y = 7$ (b) $x = -5$ (c) $2y + 9 = 0$

(d) $3x - 12 = 0$ (e) $y = 4x$ (f) $y = 0$

8. Without graphing and without solving, decide whether each parabola has **two, one, or no** x-intercepts. Justify each answer from the equation itself — not from a sketch.

(a) $y = x^2 + 5$ (b) $y = (x - 4)^2$ (c) $y = -x^2 + 9$ (d) $y = -(x + 1)^2 - 2$

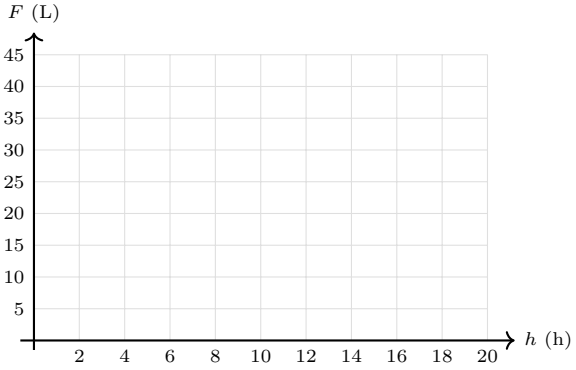
(e) Every one of the four parabolas in question 8 *does* have a y-intercept, and so does every parabola of the form $y = ax^2+bx+c$. Explain why a y-intercept can never be missing from such a graph, while an x-intercept can.

Part D — Both Intercepts Mean Something

9. A portable generator starts a job with a full fuel tank and burns fuel at a steady rate. The fuel remaining, F litres, after h hours of running is $F = 45 - 2.5h$.

(a) Complete the table of values, then plot the points and join them with a straight line.

Input (h)	Output (F)	(h, F)
0		
6		
12		
18		



(b) Determine the F-intercept algebraically and state what it represents here.

(c) Determine the h-intercept algebraically and state what it represents here.

(d) How much fuel is left after 8 hours?

(e) After how many hours is 20 L left?

(f) Both intercepts matter here, but in question 4 of the Lesson 3 assignment the n-intercept did not. What is different?

10. A small workshop's profit, P hundred dollars, on a production run of x hundred items is modelled by $P = -2x^2 + 24x - 40$, for $0 \le x \le 12$.

(a) Determine the P-intercept algebraically. **Answer as an ordered pair**, then say what it means in dollars.

(b) Factor $-2x^2 + 24x - 40$, then use it to find both x-intercepts. **Answer as ordered pairs.**

(c) What does each x-intercept represent? Give both answers in *items*.

(d) The maximum profit occurs at $x = 6$. Calculate it, in dollars.

(e) For which production runs is the workshop losing money?

Part E — Error Analysis

11. Asked to determine the **value** of the x-intercept of $2x - 5y = 20$, a student writes:

$$2(0) - 5y = 20 \;\Rightarrow\; -5y = 20 \;\Rightarrow\; y = -4, \qquad \text{so the } x\text{-intercept is } -4.$$

(a) Every line of their algebra is correct. Say exactly which substitution they should have made, and why.

(b) The number -4 is not junk — it is a real feature of this line. Which one?

(c) Determine the correct value of the x-intercept, and check it in $2x - 5y = 20$.

(d) State a one-sentence memory rule that would have stopped this mistake. Your rule must mention which *coordinate* is zero at an x-intercept, not just which letter to substitute.

12. Two students answer questions about the line $y = 3x - 12$.

(a) Asked to "state the x-intercept **as an ordered pair**," Dana writes 4. Is the number right? Is the answer right? Write what Dana should have given.

(b) Asked to "determine the **value** of the y-intercept," Emeka writes $(0, -12)$. Is the point right? Is the answer right? Write what Emeka should have given.

(c) Dana and Emeka made opposite mistakes. Describe each in one sentence, and explain why on a test the wording of the question — not the graph — decides which form is correct.

Part F — Working Backwards From the Intercepts

13. Determine the equation of the line with the given intercepts. Give each answer in the form $y = mx + b$ **and** in the form $Ax + By + C = 0$ with integer coefficients.

(a) x-intercept 6, y-intercept -4

(b) x-intercept -3, y-intercept 5

(c) x-intercept $(-2, 0)$, y-intercept $(0, -7)$

14. A line has x-intercept a and y-intercept b, where $a \neq 0$ and $b \neq 0$.

(a) Substitute $y = 0$, and then $x = 0$, into $\dfrac{x}{a} + \dfrac{y}{b} = 1$ to show that this equation really does have those intercepts.

(b) Use it to write, in one step, the equation of the line with x-intercept 5 and y-intercept -2. Clear the fractions.

(c) Why does the form fail when $a = 0$? Describe the line that causes the trouble.

15. A parabola has x-intercepts $(-2, 0)$ and $(5, 0)$ and y-intercept $(0, -20)$.

(a) Explain why $y = a(x + 2)(x - 5)$ must have the two given x-intercepts, whatever the value of a.

(b) Use the y-intercept to determine a, then write the equation in the form $y = ax^2 + bx + c$.

(c) Check all three intercepts in your expanded equation.

16. *(Multiple Choice)* Which relation has a graph with **no** x-intercept?

A. $y = x^2 - 9$ B. $y = x^2 + 9$ C. $x = 9$ D. $y = 9x$

17. *(Numerical Response)* The graph of $y = x^2 - 11x + 24$ has x-intercepts p and q, and its y-intercept has value r. The value of $p + q + r$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 4 — Extra Practice

Determining the Domain and Range of a Relation

*Throughout this set, write a domain or range with finitely many values as a **list** in braces — each value once, in increasing order (a long run of consecutive whole numbers may instead be written in set-builder notation with $\in I$) — and a domain or range with infinitely many values in **set-builder notation**, e.g. $D = \{x \mid -4 \le x \le 4,\ x \in R\}$. The variable letter must match the axis it describes.*

Part A — Lists, Tables and Mapping Diagrams

1. State the domain and range of each set of ordered pairs. Watch for values that appear more than once — a repeated value is still written only once — and put every list in increasing order.

(a) $(3,-1),\ (-2,4),\ (3,6),\ (0,-1),\ (5,4)$

(b) $(1,-3),\ (4,-3),\ (9,-3),\ (16,-3)$

(c) $(-1.5,2),\ (0.5,-3),\ (-1.5,6),\ (2.5,2)$

2. State the domain and range of the relation in each table of values.

(a)

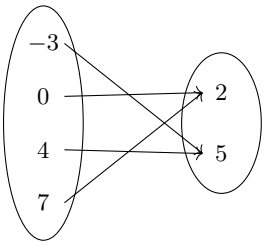
x	-4	-1	3	3	6
y	5	-2	5	8	-2

(b) The table records the height of a seedling at the end of each week.

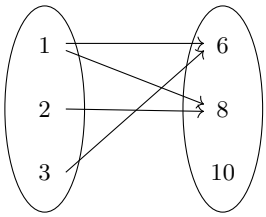
t (weeks)	0	1	2	3	4	5
h (cm)	2	5	9	9	14	20

3. State the domain and range of each mapping diagram. In part (b), look carefully: one value in the right-hand oval receives no arrow, and one value in the left-hand oval sends out two.

(a)



(b)

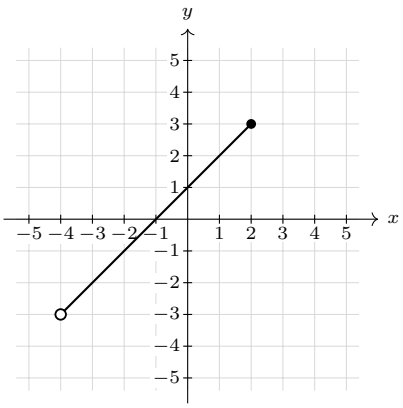


Part B — Graphs, Including Hollow Endpoints

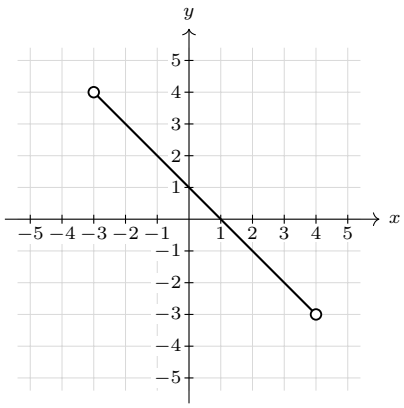
New in this set: every graphed endpoint in the booklet was *solid*. A **hollow** endpoint (an open circle) marks a point the graph gets arbitrarily close to but never reaches, so that bound is **excluded** and the inequality must be strict — $<$ or $>$ rather than $\leq$ or $\geq$.

4. State the domain and range of each relation. Each graph has at least one hollow endpoint.

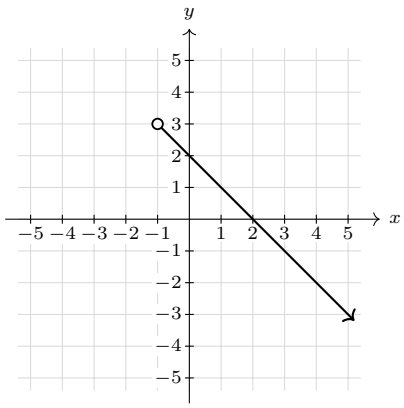
(a)



(b)

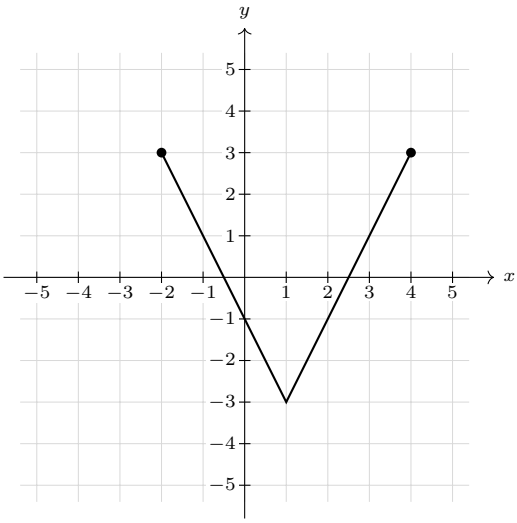


(c)

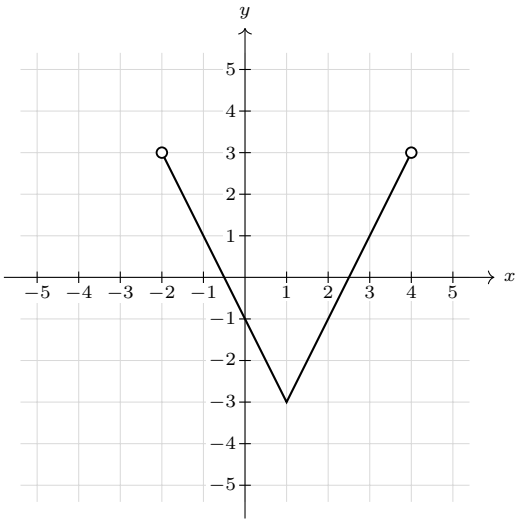


5. The two graphs below have exactly the same shape. Only the endpoints differ.

(a) State the domain and range.



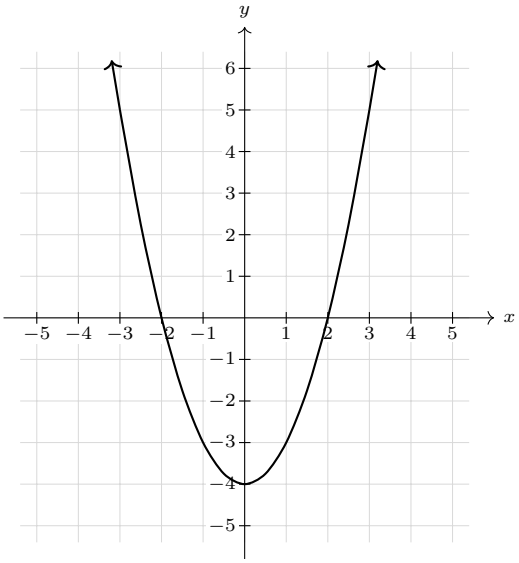
(b) State the domain and range.



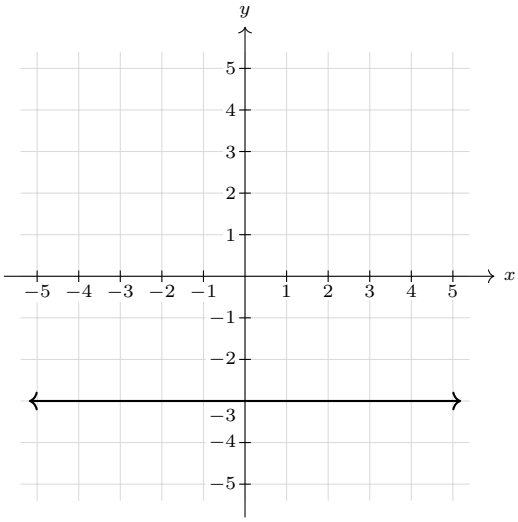
(c) Going from (a) to (b), *both* bounds of the domain changed from $\leq$ to $<$, but only *one* bound of the range did. Explain why the other bound of the range did not change.

6. State the domain and range of each curve or line below. Two of your answers are *lists*, not inequalities — decide which, and say why.

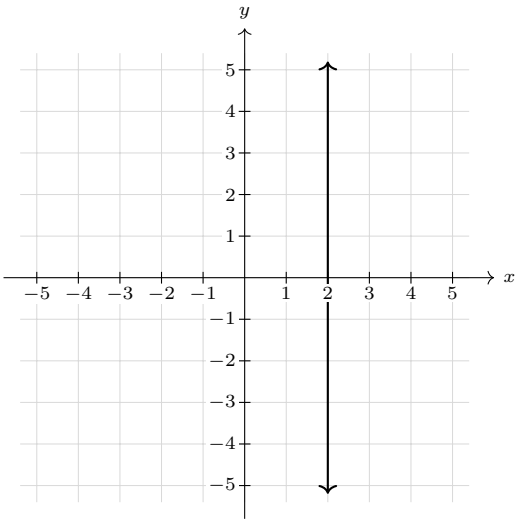
(a)



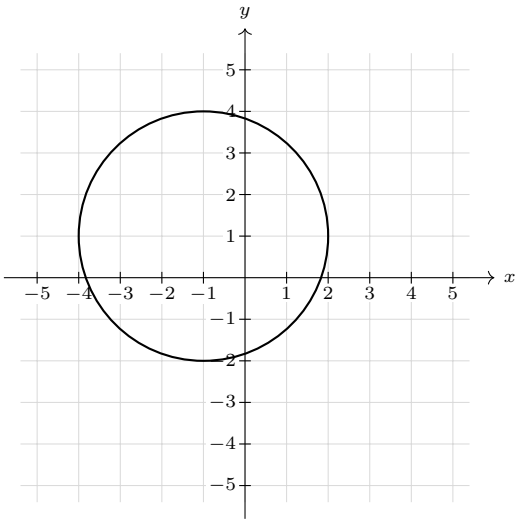
(b)



(c)



(d)



Part C — Practical Domains and Ranges

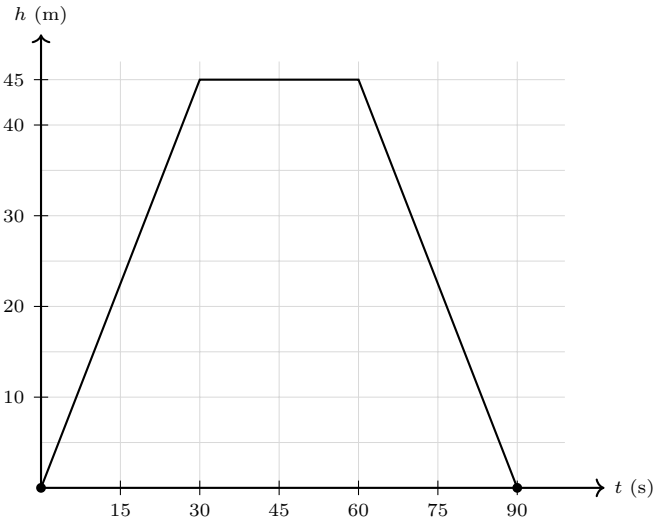
7. A drama club sells tickets to its spring play for \$12 each. The theatre seats 240 people, so the revenue R, in dollars, from selling n tickets is $R = 12n$.

(a) State the domain. Use the number set that a count of tickets belongs to, not R.

(b) A student writes $R = \{R \mid 0 \le R \le 2880,\ R \in I\}$. The two bounds are right but the notation is not. Give a value between the bounds that the relation never produces, and write the range correctly.

(c) How many ordered pairs does this relation contain?

8. A delivery drone takes off from the ground, climbs steadily to its cruising height, hovers there, and then descends to the ground. Its height h, in metres, t seconds after take-off is graphed below.



(a) State the domain and the range.

(b) Explain why the domain cannot be $\{t \mid t \in R\}$, giving one reason for the left bound and a different reason for the right bound.

(c) In question 7 the domain used $\in I$ and here it uses $\in R$. Explain the difference.

9. A 60 L rain barrel is opened and drains at a steady 4 L per minute, so the volume V of water left, in litres, after t minutes is $V = 60 - 4t$.

(a) State the domain and the range of the relation in this situation.

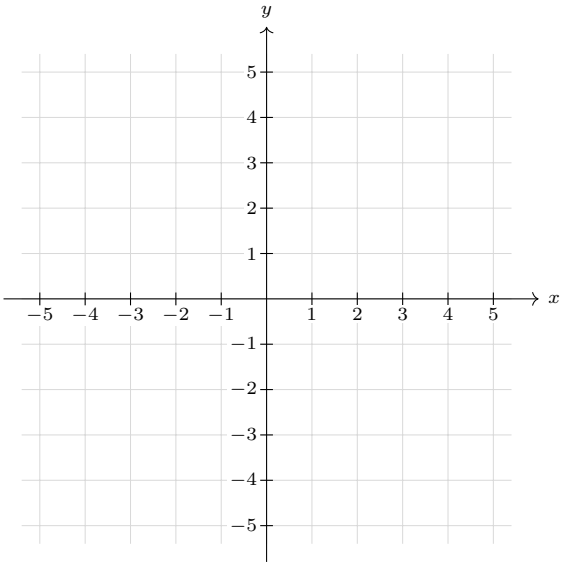
(b) A student writes $D = \{t \mid t \geq 0,\ t \in R\}$. Their lower bound is right and their upper bound is missing. What happens to the barrel that fixes the upper bound, and at what value?

(c) Now treat $V = 60 - 4t$ as a purely mathematical equation, with no barrel. State its domain and range and compare them with part (a).

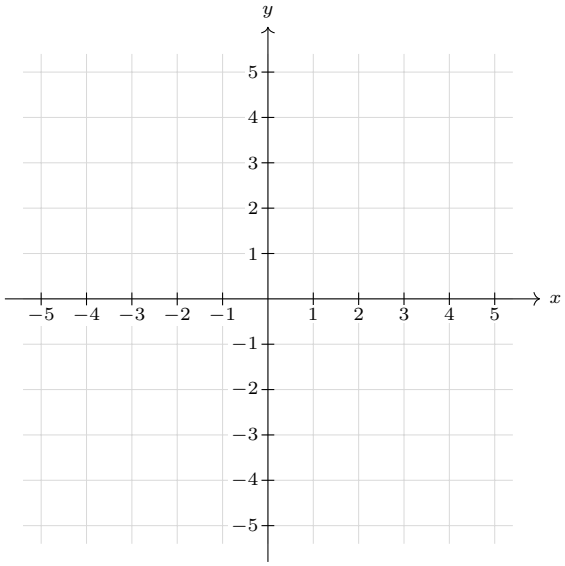
Part D — Working Backwards

10. Sketch **one** relation on each grid that has exactly the domain and range given. Mark every endpoint solid or hollow as required, and use an arrow wherever the relation continues forever.

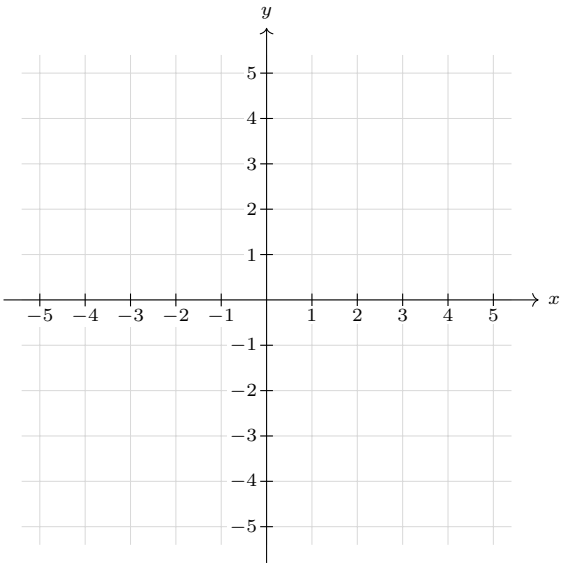
(a) $D = \{x \mid -3 \leq x \leq 4,\ x \in R\}$ and $R = \{y \mid -2 \leq y \leq 5,\ y \in R\}$



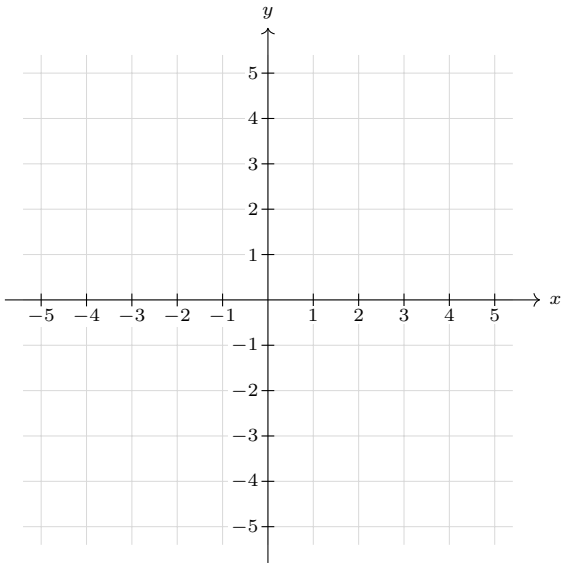
(b) $D = \{x \mid x \in R\}$ and $R = \{3\}$



(c) $D = \{-4, -1, 2, 5\}$ and $R = \{0, 3\}$



(d) $D = \{x \mid -1 < x \leq 4,\ x \in R\}$ and $R = \{y \mid -3 \leq y < 2,\ y \in R\}$



11. Answer yes or no, and give a set of ordered pairs to back up each answer.

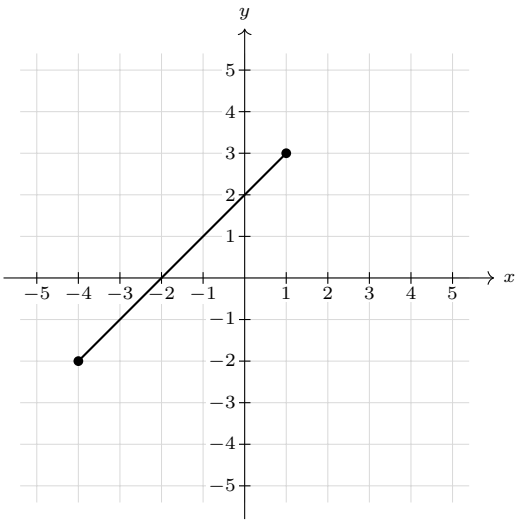
(a) Can a relation have 4 values in its domain and only 1 in its range?

(b) Can a relation have only 1 value in its domain and 4 in its range?

(c) Can a relation have 3 values in its domain and 5 in its range?

Part E — Error Analysis

12. Asked for the domain of the relation graphed below, a student writes $D = \{y \mid -2 \leq y \leq 3,\ y \in R\}$.



(a) The student has made *two* separate mistakes. Name each one.

(b) Write the correct domain and range.

13. A relation is given by $(-3, 1),\ (-1, 4),\ (2, 4),\ (5, 7)$. One student writes $R = \{y \mid 1 \le y \le 7,\ y \in R\}$. A second student writes $D = \{-3, -1, 2, 5\}$ and $R = \{1, 4, 4, 7\}$.

(a) Name a number that satisfies the first student's range but is not an output of the relation. What does that prove?

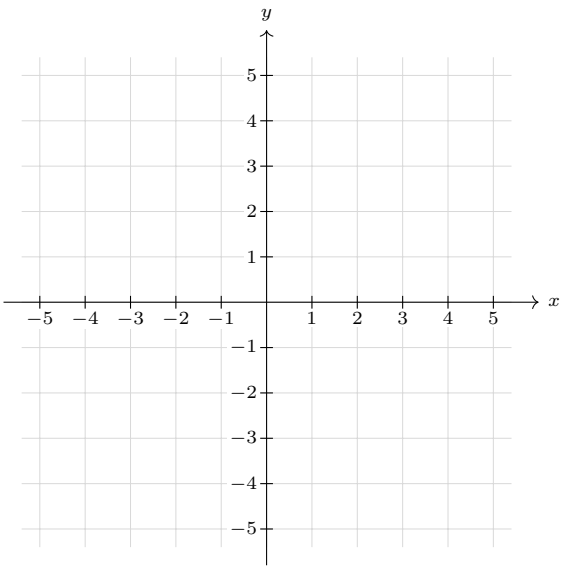
(b) The second student's range breaks one of the rules for writing a list. Which rule, and what is the correct range?

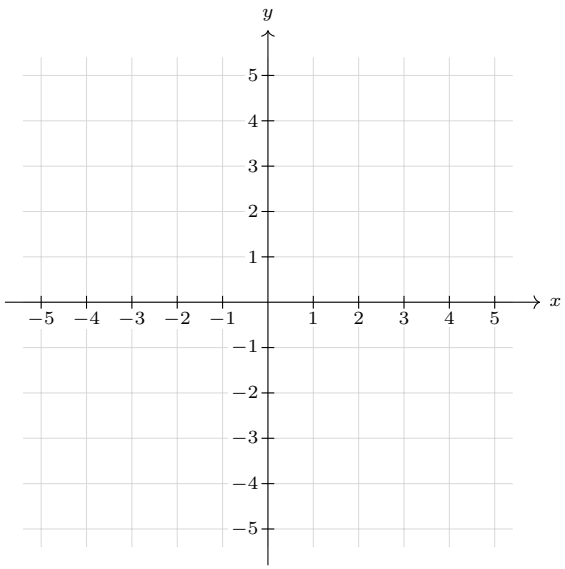
(c) State the one feature of a relation that decides whether set-builder notation is allowed at all.

Part F — A Stretch, and Two for the Test

14. Domain and range are independent of one another — fixing one of them does not fix the other.

(a) Sketch a relation on each grid below so that both have domain $\{x \mid -3 \le x \le 3,\ x \in R\}$ but their **ranges are different**. State both ranges underneath.





(b) Now describe two relations that both have range $\{y \mid -1 \le y \le 2,\ y \in R\}$ but whose **domains are different**. State both domains.

(c) A classmate says "if I tell you the domain of a relation, you can tell me its range." Use your answers above to settle the claim in one sentence.

15. True or false? If false, give a counterexample.

(a) A relation whose range is $\{5\}$ must be a horizontal line.

(b) A vertical line has exactly one value in its domain.

(c) If a graph has a hollow endpoint at $(2, 5)$, then 5 cannot be in the range.

(d) A relation with a domain of $\{x \mid x \in R\}$ must also have a range of $\{y \mid y \in R\}$.

16. *(Multiple Choice)* Which relation has domain $\{x \mid x > -2,\ x \in R\}$?
A. A ray beginning at a *solid* point $(-2, 3)$ and continuing forever to the right.
B. A ray beginning at a *hollow* point $(-2, 3)$ and continuing forever to the right.
C. The vertical line $x = -2$.
D. A parabola with vertex $(-2, 3)$ opening upward.

17. *(Numerical Response)* A circle has centre $(2, -3)$ and a radius of 7. Its domain is $\{x \mid m \leq x \leq n,\ x \in R\}$ and its range is $\{y \mid p \leq y \leq q,\ y \in R\}$. The value of $n + q$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 5 — Extra Practice

Functions and Function Notation

One test runs through this whole set: a relation is a ***function*** *when every element of the domain is paired with* ***exactly one*** *element of the range. Nothing in that sentence restricts the outputs — two different inputs are free to share an output. Once a relation is known to be a function, $f(x)$ names its* output value *at the input x, while f on its own names the* function*; $f(3) = 7$ and "the point $(3, 7)$ lies on the graph of $y = f(x)$" say the same thing.*

Part A — The Function Test in Five Representations

1. State whether each set of ordered pairs is a function. Give the reason — if it is not a function, name the input that breaks the test and the outputs it is paired with.

(a) $(-4, 3),\ (-1, 3),\ (2, 3),\ (5, 3)$

(b) $(6, 1),\ (6, 2),\ (6, 3),\ (6, 4)$

(c) $(0, -2),\ (3, 5),\ (-3, 5),\ (0, 2)$

(d) $(1.5, 4),\ (-1.5, 4),\ (4, 1.5),\ (-4, 1.5)$

2. State whether the relation in each table is a function, and say why.

(a)

x	-3	0	2	5	8
y	7	7	7	7	7

(b)

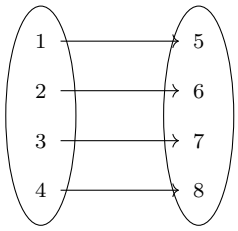
t	0	1	2	2	3
d	0	4	8	12	16

(c)

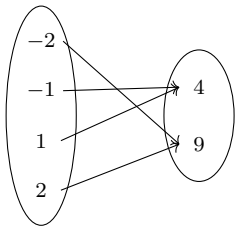
n	1	2	3	4
C	12	24	12	24

3. State whether each mapping diagram shows a function. Count *arrows leaving each left-hand value* — that, and nothing else, is the test.

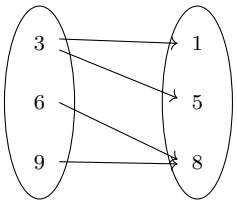
(a)



(b)

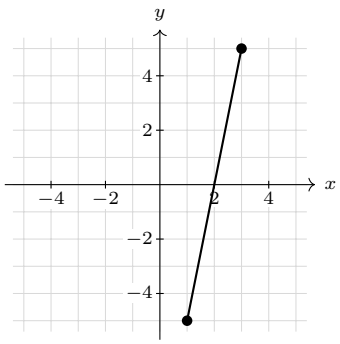


(c)

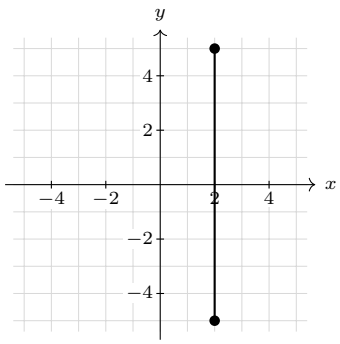


4. Apply the vertical line test to each graph. Graphs (a) and (b) are deliberately similar — use the grid, not your first impression, to tell them apart.

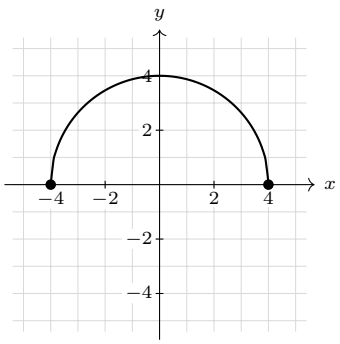
(a)



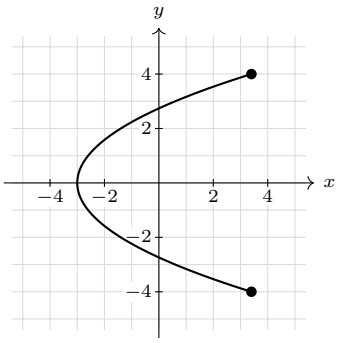
(b)



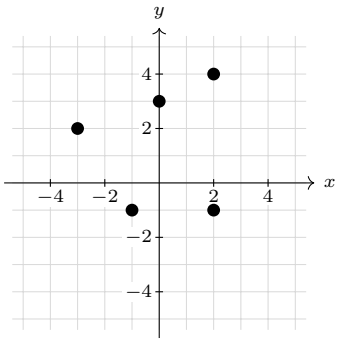
(c)



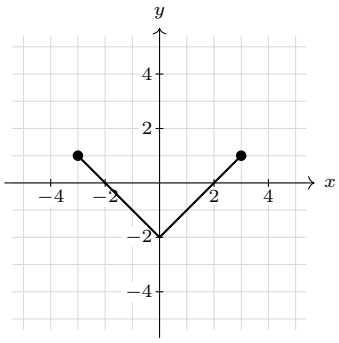
(d)



(e)



(f)



5. Decide whether each equation defines y as a function of x. When it does not, give one x-value together with two different y-values it produces.

(a) $y = 2x - 9$

(b) $y = x^2 - 6x$

(c) $x = y^2 + 1$

(d) $y = 7$

(e) $x = 7$

(f) $x^2 + y^2 = 36$

6. Parts (b) and (c) of question 5 are closely related: in one the input is squared, in the other the output is.

(a) $y = x^2$ is a function but $x = y^2$ is not, even though one is obtained from the other by swapping the letters. Explain what the swap does to the function test.

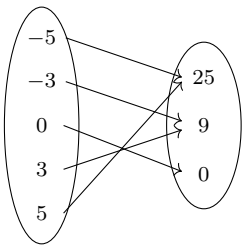
(b) Explain why $y = \sqrt{x}$ *is* a function of x while $y^2 = x$ is not. (The $\sqrt{\ }$ symbol means the *principal* square root only.)

Part B — One-to-One, Many-to-One, One-to-Many

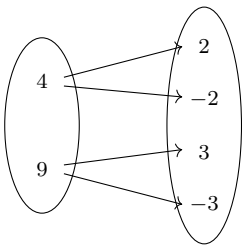
New in this set. A relation is **one-to-one** when each input has one output *and* each output comes from one input; **many-to-one** when two or more inputs share an output; **one-to-many** when a single input produces two or more outputs. The first two are functions. Only the third fails the test.

7. For each mapping diagram, state whether it is one-to-one, many-to-one or one-to-many, and then state whether it is a function.

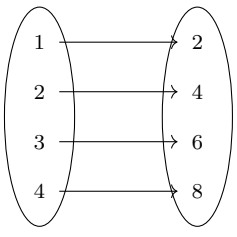
(a)



(b)



(c)



8. Classify each set of ordered pairs as one-to-one, many-to-one or one-to-many, then state whether it is a function.

(a) $(1,3),\ (2,6),\ (3,9),\ (4,12)$

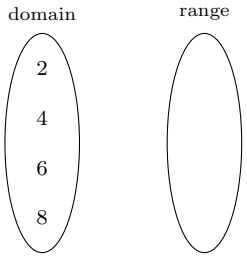
(b) $(1,5),\ (2,5),\ (3,5),\ (4,5)$

(c) $(1,2),\ (1,4),\ (2,6),\ (3,8)$

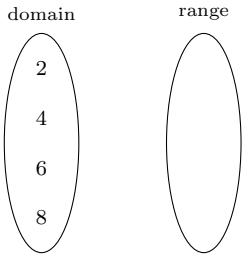
(d) $(-2,4),\ (2,4),\ (-3,9),\ (3,9)$

9. Draw your own mapping diagram in each oval pair below. Every diagram must use the domain $\{2,4,6,8\}$.

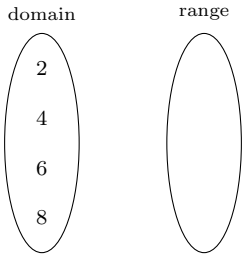
(a) many-to-one, and a function



(b) one-to-one



(c) one-to-many



(d) Exactly one of your three diagrams fails the function test. Say which, and quote the definition of a function that it breaks.

(e) Is it possible for a relation to be both one-to-one and one-to-many at the same time? Explain.

Part C — Function Notation: the Name, the Value, and the Graph

10. Each line below says something about a function. Rewrite it in the form asked for. Take $y = f(x)$ to be the equation of the graph of f.

(a) $f(2) = -5$. Write the ordered pair this puts on the graph of f.

(b) The point $(-3, 8)$ lies on the graph of f. Write this in function notation.

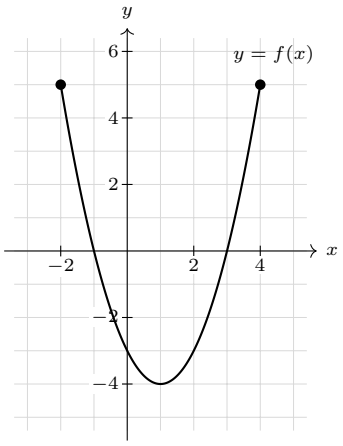
(c) The image of -1 under f is 6. Write this in function notation and as a point.

(d) $g(0) = 4$. Name the feature of the graph of g this describes.

(e) $h(5) = 0$. Name the feature of the graph of h this describes.

(f) $f(4) = f(-4)$. Describe, in words, what this says about the graph of f.

11. The graph of a function f is shown. Read every answer off the graph.



(a) $f(0)$

(b) $f(3)$

(c) All values of x for which $f(x) = 5$.

(d) All values of x for which $f(x) = -3$.

(e) The equation $f(x) = -4$ has exactly one solution. State it, and explain what is special about that point of the graph.

(f) The equation $f(x) = -6$ has no solution. Explain how the graph shows this, and why it does *not* mean f fails to be a function.

12. A function f is given entirely by the table below.

x	-4	-2	0	1	3	6
$f(x)$	7	1	-3	1	5	7

(a) State $f(-2)$ and $f(3)$.

(b) Solve $f(x) = 7$.

(c) Solve $f(x) = 1$.

(d) Explain why the repeated output 7 does not stop f from being a function.

(e) What is the answer to "solve $f(x) = 0$" for this function?

(f) Write the six ordered pairs that lie on the graph of $y = f(x)$.

Part D — Evaluating: Numbers and Expressions as Inputs

13. A function is defined by $f(x) = x^2 - 4x + 1$. Determine each of the following, simplified.

(a) $f(-2)$

(b) $f(a)$

(c) $f(2a)$

(d) $f(a + 1)$

(e) $f(a) + 1$

(f) Your answers to (d) and (e) are different. Explain, in words, why the $+1$ lands in a different place.

14. A function is defined by $g(x) = 5 - 2x$. Determine each of the following, simplified.

(a) $g(-4)$ (b) $g(2a)$ (c) $g(a + 1)$ (d) $g(a) - g(a + 1)$

(e) Your answer to (d) contains no a. Explain what that tells you about the graph of g.

(f) Determine $g(a) + g(1)$ and compare it with your answer to (c). Is there any value of a for which $g(a + 1) = g(a) + g(1)$? Justify your answer.

15. Two functions are defined by $f(x) = x^2 + 1$ and $g(x) = 2x + 4$.

(a) $f(2) + g(2)$

(b) $f(-1) \cdot g(0)$

(c) Evaluate $g(3)$, then use that answer as the input to f.

(d) Solve $f(x) = g(x)$.

16. For a function f, the expression $f(a+1) - f(a)$ measures how much the output changes when the input goes up by 1. Determine and simplify $f(a+1) - f(a)$ for each function.

(a) $f(x) = 3x - 5$

(b) $f(x) = x^2$

(c) $f(x) = x^2 - 4x + 1$

(d) One of your three answers does not depend on a. Which type of function was it, and why does that happen?

Part E — Working Backwards: Given the Output, Find the Input

17. Solve for the input in each case. Two of the six behave differently from the rest — say so when they do.

(a) $f(x) = 7 - 4x$; solve $f(x) = -13$.

(b) $g(x) = \dfrac{x}{3} + 8$; solve $g(x) = 2$.

(c) $h(x) = x^2 - 5$; solve $h(x) = 44$.

(d) $P(x) = 2x^2 + 3$; solve $P(x) = 3$.

(e) $k(x) = 50 - 2x^2$; solve $k(x) = 68$.

(f) $m(x) = \sqrt{x} + 4$; solve $m(x) = 11$.

18. A function is defined by $f(x) = 3x - 8$. Solve each equation for x.

(a) $f(x) = x$

(b) $f(x) = 0$

(c) $f(2x) = 10$

(d) $f(x + 3) = 1$

Part F — Find the Mistake

19. *(Error analysis)* A function is defined by $f(x) = 4x + 5$. A student computes $f(1) = 9$, then writes:

"$f(3)$ means f times 3, and f is 9, so $f(3) = 27$."

(a) Calculate $f(3)$ correctly.

(b) The student has made *two* separate mistakes: one about what the brackets mean, one about what the letter f stands for. Name both.

(c) The same student writes $f(x + 2) = f(x) + 2$. Test this with $f(x) = x^2$ and $x = 1$, and state a correct expression for $f(x + 2)$.

(d) Show that for the function $f(x) = 4x$ the student's claim $f(3) = 3f(1)$ happens to give the right number. Explain why a right answer here is still a wrong method.

(e) Write the meaning of $f(3)$ in one sentence that could not be misread as multiplication.

20. *(Error analysis)* Asked whether $\{(-2, 9),\ (-1, 4),\ (1, 4),\ (2, 9)\}$ is a function, a student answers: "No. The output 4 is used twice and the output 9 is used twice, so the outputs are not unique."

(a) State the definition of a function, then say which *side* of each ordered pair it restricts.

(b) Classify the relation as one-to-one, many-to-one or one-to-many, and give the correct verdict.

(c) Change one ordered pair so that the relation really is not a function. State your new set and the input that breaks the test.

Part G — Reasoning About the Definition

21. Answer each question with a sentence, and support it with an example or a counterexample.

(a) Can a function have two *different inputs* that produce the *same output*? If so, give a set of ordered pairs; if not, explain why not.

(b) Can a function have two *different outputs* for the *same input*? If so, give a set of ordered pairs; if not, explain why not.

(c) Every point on the vertical line $x = c$ shares the same x-coordinate. Use that one fact to explain why the vertical line test is exactly the definition of a function in disguise.

(d) A student proposes a "horizontal line test" for functions. Explain why it does *not* decide whether a relation is a function, and say what it does decide instead.

22. State whether each statement is true or false. Correct every false statement.

(a) $f(x)$ means f multiplied by x.

(b) If $f(2) = 9$, the point $(2, 9)$ lies on the graph of $y = f(x)$.

(c) If $f(2) = 9$ and $f(7) = 9$, then f is not a function.

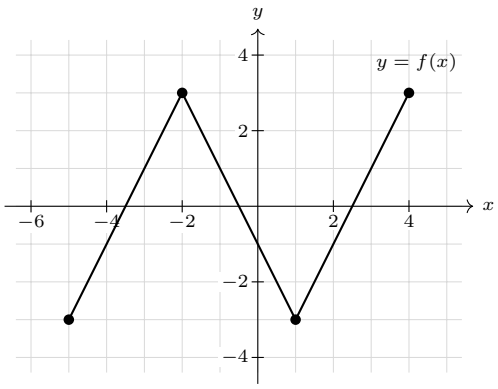
(d) If $f(2) = 9$ and $f(2) = 5$, then f is not a function.

(e) f and $f(x)$ mean the same thing.

(f) $f(a + b) = f(a) + f(b)$ for every function f.

Part H — A Stretch, and Two for the Test

23. *(Stretch)* The graph of a function f is made of three straight pieces joined end to end, through the points $(-5,-3)$, $(-2,3)$, $(1,-3)$ and $(4,3)$.



(a) State $f(-2)$, $f(0)$ and $f(3)$.

(b) Solve $f(x) = 3$.

(c) Solve $f(x) = -3$.

(d) Find every value of a with $a \neq 3$ for which $f(a) = f(3)$.

(e) Explain why this graph is a function even though the equation $f(x) = 1$ has three solutions.

(f) For which values of k does $f(x) = k$ have exactly **three** solutions? Give your answer as an inequality.

24. *(Multiple Choice)* A function f satisfies $f(-2) = 9$. Which statement must be true?
A. The point $(9,-2)$ lies on the graph of $y = f(x)$.
B. The point $(-2,9)$ lies on the graph of $y = f(x)$.
C. f multiplied by -2 equals 9.
D. $f(9) = -2$.

25. *(Numerical Response)* A function is defined by $f(x) = x^2 + 5x$. The equation $f(x) = 104$ has two solutions, p and q, with $p > q$. The value of $p - q$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 7 — Extra Practice

Function Notation: Graphs, Conversions, and Real-World Models

*Two readings run through this whole set, and they are opposites. $f(3)$ hands you the **input** 3 and asks for the output: start on the horizontal axis, read up or down to the graph, and there is **exactly one** answer. $f(x) = 3$ hands you the **output** 3 and asks for the input: start on the vertical axis, read across, and count* every *point of the graph at that height — there may be one, several, a whole interval, or none. A third idea joins them: in a real-world model, $f(a) = b$ is a sentence about the situation, so its interpretation is written as a sentence, with units, not as a bare number.*

Part A — Two Notations, One Rule

1. Express each relation given in function notation as an equation in two variables. Watch the letter: the independent variable never changes when you convert.

(a) $f(x) = 9 - 4x$ (b) $g(t) = t^2 - 5t$ (c) $P(n) = 0.5n + 3.25$ (d) $h(a) = \dfrac{2a - 7}{3}$

2. Express each relation in function notation. Three of these must be rearranged first, and one of them cannot be done at all — say which, and why.

(a) $y = 7 - \dfrac{w}{2}$ (b) $y = 6x^2 - x$ (c) $5x - 2y = 14$

(d) $4x + 3y + 18 = 0$ (e) $2y - 9 = 0$ (f) $x = 5$

3. Each table below gives every ordered pair of a function. Write the rule first in x-y notation and then in function notation, keeping the letters used in the table.

(a)

x	-2	-1	0	1	2
y	11	8	5	2	-1

(b)

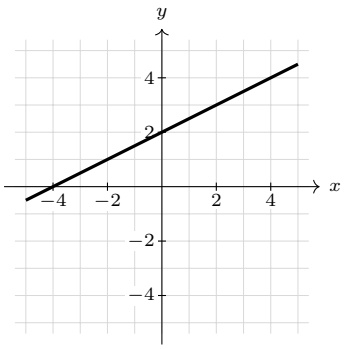
t	0	2	4	6
d	7	16	25	34

(c)

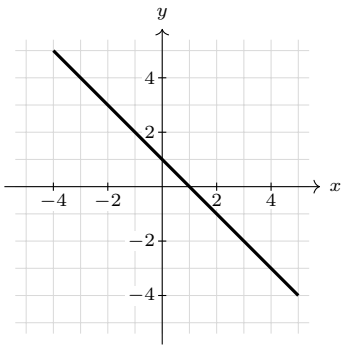
n	1	2	3	4
A	1	4	9	16

4. Each graph below is the graph of a function f. Write its equation in x-y notation and then in function notation. Read two lattice points off each graph first.

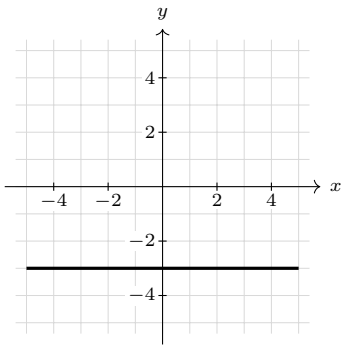
(a)



(b)



(c)



5. Each conversion below is wrong. Say what was changed that should not have been, and write the correct version.

(a) "$h(t) = t^2 + 3t - 8$ becomes $y = x^2 + 3x - 8$."

(b) "$y = 9x - 4$ becomes $f(y) = 9x - 4$."

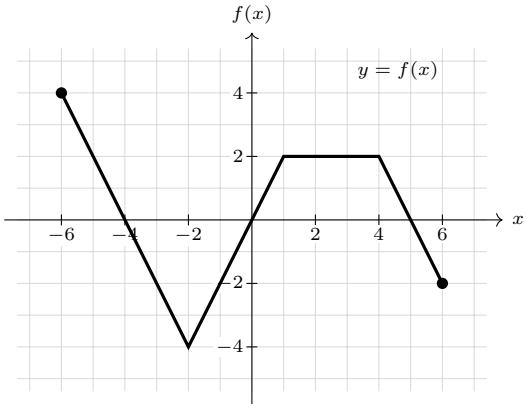
(c) "$3x + 4y = 24$ becomes $f(x) = 24 - 3x - 4$."

(d) "$f(x) = 2x - 3$ becomes $y = 2y - 3$."

(e) In one sentence, state what changes and what does *not* change when a rule is rewritten in function notation.

Part B — Reading a Graph in Both Directions

6. The graph of a function f is shown. Read every answer off the graph, and count *every* point at the height you are given.



(a) State $f(-5)$, $f(-3)$, $f(0)$, $f(3)$ and $f(6)$.

(b) Solve $f(x) = 0$.

(c) Solve $f(x) = 4$.

(d) Solve $f(x) = -5$, and explain how the graph tells you so without any algebra.

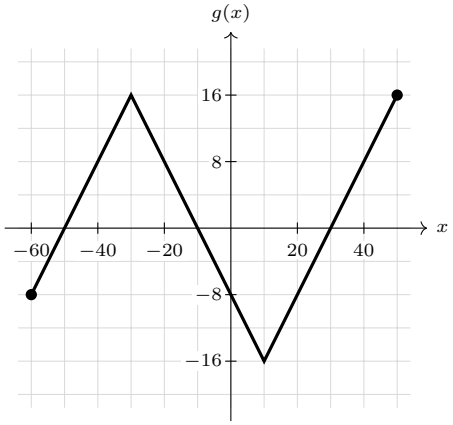
(e) Solve $f(x) = 2$. This answer is not a short list — describe it exactly.

(f) Write the x-intercepts and the y-intercept of the graph using function notation.

(g) State the domain and the range of f.

(h) Your answers to (a) each contained one number, while (b), (c), (e) contained different amounts. Explain, using the definition of a function, why that had to happen.

7. The graph of a function g is shown below. *Each grid line represents* 10 *units on the horizontal axis and* 4 *units on the vertical axis.* Check the scale before every read.



(a) State $g(-60)$, $g(-40)$, $g(0)$ and $g(20)$.

(b) Write the ordered pairs belonging to the four values in (a).

(c) Solve $g(x) = 8$.

(d) Solve $g(x) = 16$.

(e) Solve $g(x) = -20$, and say what the graph shows.

(f) Write the x-intercepts of the graph in function notation, and state the domain and range of g.

(g) A student read $g(-3) = 4$ straight off the grid squares. Explain the mistake and give the two correct statements that the point actually supports.

8. No graph is given. For each item, state (i) whether you have been given an input or an output, (ii) which axis you would start from, and (iii) how many answers are possible.

(a) Find $f(6)$.

(b) Solve $f(x) = 6$.

(c) Find $f(0)$.

(d) Solve $f(x) = 0$.

(e) Given $f(k) = -2$, find k.

(f) Given $f(-2) = k$, find k.

(g) Explain why "find $f(6)$" can never produce two answers, no matter what the function is, while "solve $f(x) = 6$" often does.

Part C — Models in Function Notation

9. A courier charges by mass. The cost of shipping a parcel is modelled by $C(m) = 8.5m + 22$, where $C(m)$ is the cost, in dollars, of shipping a parcel of mass m kilograms.

(a) Determine $C(12)$.

(b) Write the meaning of your answer to (a) as a sentence, with units.

(c) Determine $C(0)$ and say what it represents.

(d) Solve $C(m) = 90$, and write the meaning of the solution as a sentence.

(e) Solve $C(m) = 15$. Interpret the result in the context of the parcels.

(f) "$C(90)$" and "$C(m) = 90$" are different questions. State in words what each one asks.

10. In a deep mountain lake the water temperature drops steadily with depth. It is modelled by $T(d) = 21 - 0.4d$, where $T(d)$ is the temperature, in degrees Celsius, at a depth of d metres. The lake is 40 m deep at its deepest point.

(a) Determine $T(25)$ and write its meaning as a sentence.

(b) Determine $T(0)$ and say what feature of the graph it is.

(c) Solve $T(d) = 5$ and write the meaning of the solution as a sentence.

(d) Solve $T(d) = 25$. Explain why the number you get is not an answer to the question asked.

(e) Solve $T(d) = 3$. Explain why this has no answer in this lake even though the equation has a solution.

(f) State a domain and a range for T that fit the lake.

11. A ski hill records $V(t)$, the number of visitors on day t of a six-day festival. The function is given entirely by the table below — there is no formula.

t (day)	1	2	3	4	5	6
$V(t)$ (visitors)	240	310	380	310	240	450

(a) State $V(3)$, and write its meaning as a sentence.

(b) Solve $V(t) = 310$ and write the meaning of your answer as a sentence.

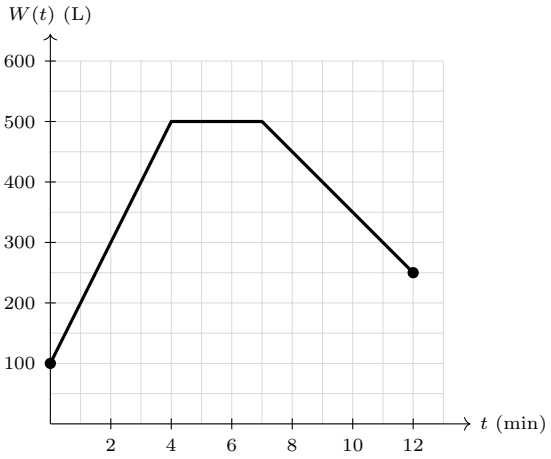
(c) Solve $V(t) = 300$.

(d) Explain why the repeated output 240 does not stop V from being a function.

(e) State the domain and range of V, using the correct membership set for a count of days.

(f) Would it be reasonable to join the six points with a line? Give a reason.

12. A holding tank is filled, held, then drained. The graph shows $W(t)$, the volume of water in the tank, in litres, t minutes after filling begins.



(a) State $W(0)$, $W(3)$, $W(6)$ and $W(10)$.

(b) Write the meaning of $W(9) = 400$ as a sentence, with units.

(c) Solve $W(t) = 400$, then write the meaning of *both* solutions as one sentence.

(d) Solve $W(t) = 500$. Describe the answer exactly, and say what the tank is doing then.

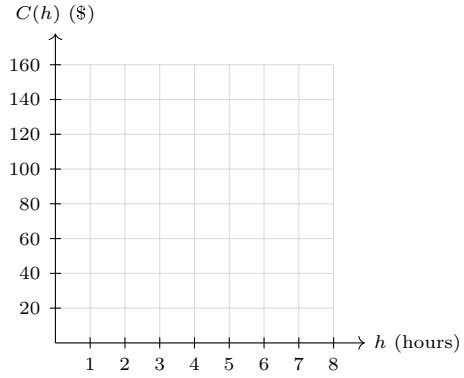
(e) Solve $W(t) = 600$ and $W(t) = 100$. One has no answer and one has exactly one — explain both.

(f) State the domain and range of W.

(g) Is the tank filling faster than it drains, or the other way round? Support your answer with two values read from the graph.

13. A guide service rents sea kayaks. The charge is modelled by $C(h) = 15h + 40$, where $C(h)$ is the cost, in dollars, of a rental lasting h hours. Rentals are limited to 8 hours.

(a) Sketch the graph on the grid provided.



(b) Determine $C(3)$, and write its meaning as a sentence with units.

(c) Determine $C(0)$ and explain what it represents about the rental.

(d) Solve $C(h) = 115$ and write the meaning of the solution as a sentence.

(e) Solve $C(h) = 200$. Explain why the guide service cannot fill this order.

(f) State the domain and range of C for this rental.

14. Write the meaning of each statement as a **sentence**, with units. A number on its own is not an answer to any part of this question.

(a) $P(n)$ is a bakery's profit, in dollars, from selling n loaves, and $P(140) = 95$.

(b) For the same bakery, the equation $P(n) = 0$ has the solution $n = 60$.

(c) $D(t)$ is a driver's distance from home, in kilometres, t hours into a trip, and $D(2) = D(5)$.

(d) $F(c)$ is the cost, in dollars, of c cups of coffee, and solving $F(c) = 12$ gives $c = 3$.

(e) $H(t)$ is a drone's height, in metres, t seconds after launch, and $H(t) = 20$ has the two solutions $t = 4$ and $t = 11$.

(f) For the same drone, $H(t) = 65$ has no solution.

Part D — Find the Mistake

15. *(Error analysis)* A function is defined by $f(x) = 3x - 4$. Asked to **solve** $f(x) = 5$, a student writes:

"$f(5) = 3(5) - 4 = 11$, so the answer is 11."

(a) Identify the student's error in one sentence: which number was treated as an input, and which one should have been?

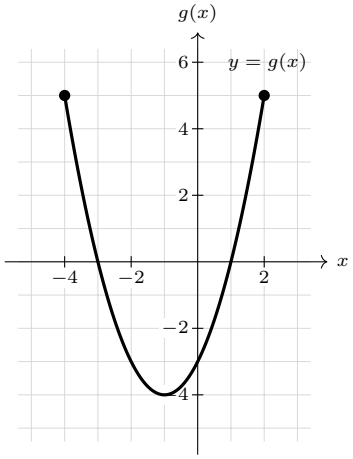
(b) Solve $f(x) = 5$ correctly.

(c) State the question that "11" is the correct answer to, and write it in function notation.

(d) For the function $f(x) = x$, the student's method gives the right answer to "solve $f(x) = 5$". Show that it does, then explain why a right answer here does not rescue the method.

(e) On a graph, one of "$f(5)$" and "$f(x) = 5$" is found by starting on the horizontal axis and one by starting on the vertical axis. Say which is which, and state how many answers each can have.

16. *(Error analysis)* The graph of a function g is shown. Asked to solve $g(x) = 0$, a student answers "$x = 1$". Asked to solve $g(x) = 5$, the same student answers "$x = 2$".



(a) Give the complete solution to $g(x) = 0$ and the complete solution to $g(x) = 5$.

(b) Describe, as a rule the student could follow, what to do after finding the first crossing.

(c) A second student sees your answer to (a) and says "two different x-values give the same y-value, so g is not a function." Correct this.

(d) State a value of b for which $g(x) = b$ has exactly one solution, and a value for which it has none.

(e) State $g(0)$ and $g(-1)$, and explain why neither of these reads could ever have produced two answers.

17. *(Error analysis)* A market sells apples by mass, and $C(k)$ is the cost, in dollars, of k kilograms. It is known that $C(12) = 47$. Three students interpret this.

(a) Ava writes: "\$12 buys 47 kg of apples." Explain her error and write the correct sentence.

(b) Ben writes: "the point $(47, 12)$ lies on the graph of $y = C(k)$." Explain his error and write the correct point.

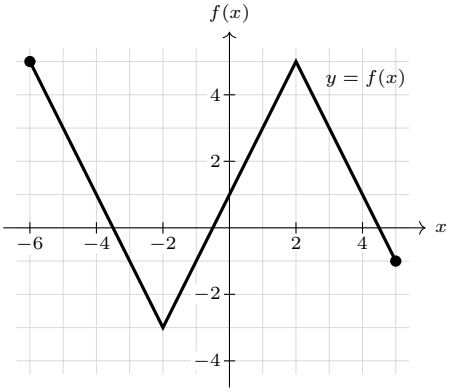
(c) Cam writes: "C multiplied by 12 is 47." Explain his error in terms of what the letter C names.

(d) Write, in function notation, the statement "8 kg of apples cost \$32", and then write the question "how many kilograms can be bought for \$32?" in function notation as well.

(e) Explain why $C(12) = 47$ and $C(47) = 12$ are different claims, and describe a function (not necessarily a sensible price list) for which both could be true.

Part E — A Stretch, and Three for the Test

18. *(Stretch)* The graph of a function f is made of three straight pieces joined end to end, through $(-6, 5)$, $(-2, -3)$, $(2, 5)$ and $(5, -1)$.



(a) State $f(-4)$, $f(0)$ and $f(4)$, and write what you notice in function notation.

(b) Solve $f(x) = 1$ and $f(x) = 5$.

(c) Solve $f(x) = -2$. There are two answers and neither lands on a grid line.

(d) Solve $f(x) = 6$ and $f(x) = -3$, and say how many answers each has.

(e) For which values of b does $f(x) = b$ have exactly **three** solutions?

(f) State every value of b for which $f(x) = b$ has exactly **two** solutions. Your answer has two parts — check the top and the bottom of every piece before you write it down.

(g) The three pieces do not all reach the same lowest height. Explain how that single fact is what creates the interval of b-values in (f), and predict what would change if the last piece ended at $(5, -3)$ instead.

19. *(Multiple Choice)* A courier's cost function satisfies $C(4) = 31$, where $C(m)$ is the cost, in dollars, of shipping a parcel of mass m kilograms. Which statement must be true?
A. A parcel costing \$4 has a mass of 31 kg.
B. Shipping a 4 kg parcel costs \$31.
C. C multiplied by 4 equals 31.
D. $C(31) = 4$.

20. *(Multiple Choice)* For the function f graphed in question 6, the equation $f(x) = 2$ has A. one solution B. two solutions C. three solutions D. infinitely many solutions

21. *(Numerical Response)* A function is defined by $f(x) = x^2 - 6x$. The equation $f(x) = 27$ has two solutions, p and q, with $p > q$. The value of $p - q$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 8 — Extra Practice

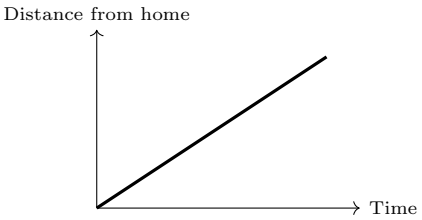
Interpreting and Sketching Real-World Graphs

*A graph of a real-world situation tells a story, and every part of the story is carried by the **shape**: rising, falling, flat, straight, bending. Two questions run through this whole set. First,* what is on the vertical axis? *— a flat segment means something completely different on a distance-time graph than on a speed-time graph. Second,* how steep is it? *— steepness is the rate of change, and it carries units.*

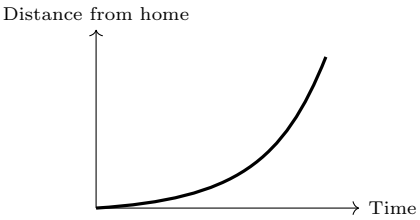
Part A — Matching a Story to Its Graph

1. Each graph below shows one person's distance from home plotted against time. Match each story with the graph that fits it, and give a **reason** — name the feature of the graph that settles it. Graphs **A** and **B** begin and end at exactly the same points, so only what happens *in between* can tell them apart.

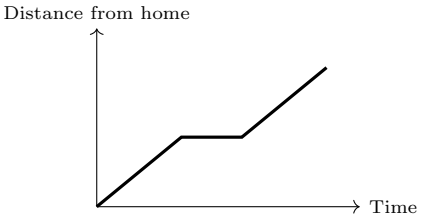
A.



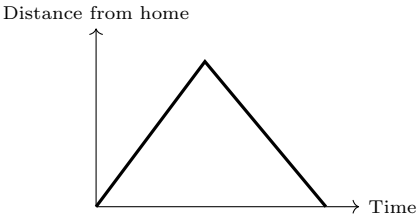
B.



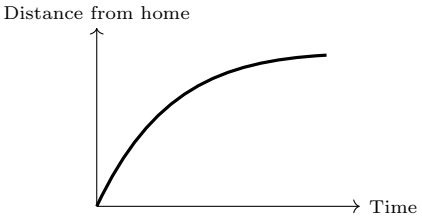
C.



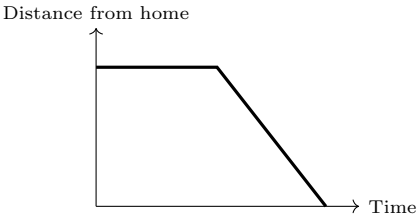
D.



E.



F.



(a) Amir jogs from home to the soccer pitch at one steady pace and never stops.
Graph: _____ Reason:

(b) Beth wheels her bike out of the driveway and rides away, pedalling harder and harder, so that each minute she covers more ground than the minute before.
Graph: _____ Reason:

(c) Cai drives away from home, is held at a level crossing while a train passes, then carries on at the speed he was going before.
Graph: _____ Reason:

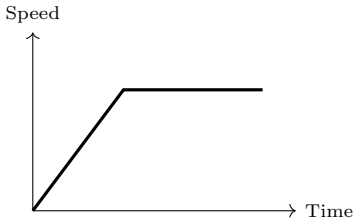
(d) Dev walks to the corner store and comes straight back home without going inside.
Graph: _____ Reason:

(e) Eryn's train pulls out of her home station and then slows steadily as it coasts into the next station down the line.
Graph: _____ Reason:

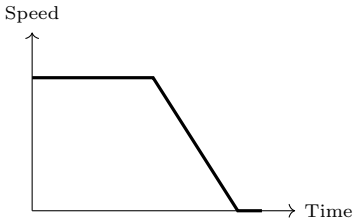
(f) Fen is already at the library when the graph starts. She studies there for an hour, then walks home.
Graph: _____ Reason:

2. These four graphs show **speed** against time — read the vertical axis before you decide. Match each story with its graph and give the reason.

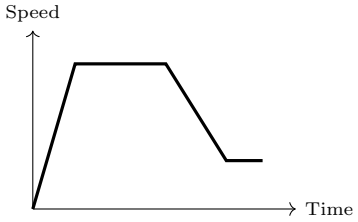
P.



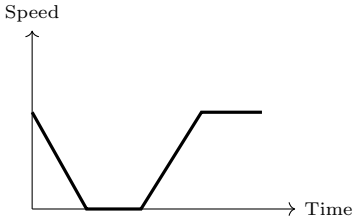
Q.



R.



S.



(a) A city bus pulls away from a stop, builds up to the speed limit, and holds it all the way to the next stop.
Graph: _____ Reason:

(b) A car is cruising on the highway when the driver brakes smoothly to a complete stop at a rest area.
Graph: _____ Reason:

(c) A car driving through town slows to a stop at a red light, waits for it to change, then builds back up to the speed it was doing before.
Graph: _____ Reason:

(d) A sprinter accelerates hard out of the blocks, holds top speed to the finish line, then slows to an easy jog and keeps jogging.
Graph: _____ Reason:

3. Look again at graphs **A** and **B** from question 1. Suppose both riders leave home at the same instant, both graphs start at (0 min, 0 km), and both end at (30 min, 6 km).

(a) Find the average speed of each rider over the whole 30 minutes, in km/h. What do you notice?

(b) Which rider is moving faster in the first few minutes? Which is moving faster in the last few? Say how the graph tells you.

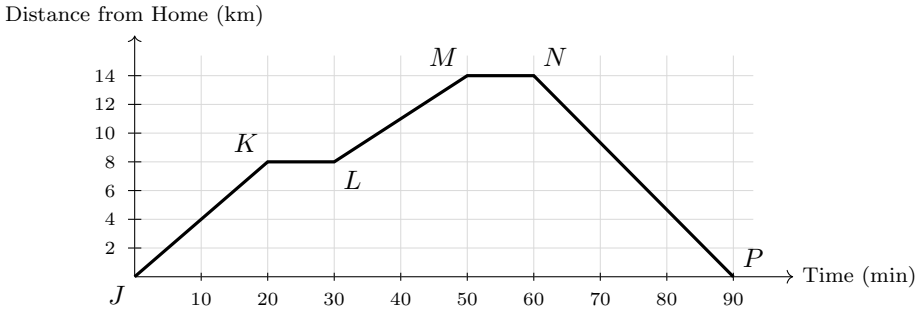
(c) Which rider is further from home 15 minutes in? Explain using the position of B's curve relative to A's line.

(d) B starts out slower than A and finishes faster than A. Explain why there must be at least one instant during the ride when the two riders are travelling at exactly the same speed, and say what that speed is.

(e) "Equal average speed means equal speed the whole way." Use these two graphs to show that the statement is false.

Part B — Describing a Graph, Segment by Segment

4. Jules rides her bike from home to a friend's house, then rides home again. The graph of her distance from home is made of five straight segments meeting at J, K, L, M, N, P.



(a) Describe what Jules is doing on each segment. Use one of the words *steady*, *stopped* or *returning* in every answer, and give the distance and the time involved.
(i) J to K (ii) K to L (iii) L to M (iv) M to N (v) N to P

(b) How far from home is the friend's house, and how long does Jules spend there?

(c) For how many minutes in total is Jules not moving? Name every part of the graph you used.

(d) How far does Jules travel altogether? Why is that not the same as her distance from home at P?

5. Rates of change from the same graph. Give every answer **with its units**.
(a) Find the rate of change on J to K, first in km/min and then in km/h.

(b) Find the rate of change on L to M in km/h. Is Jules faster before her first stop or after it, and how does the picture show that?

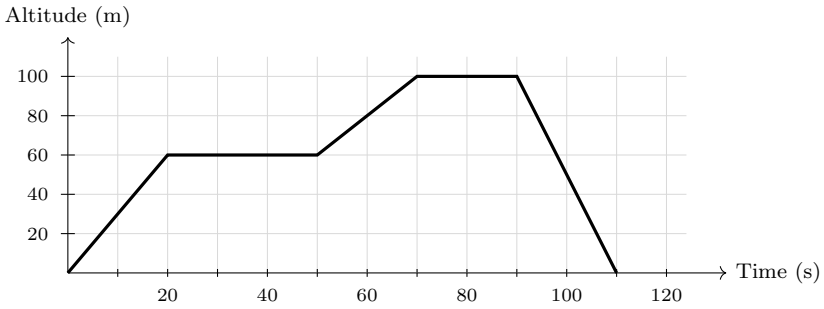
(c) Find the rate of change on K to L in km/h, and say what that number means about Jules.

(d) Find the rate of change on N to P in km/h. Explain what the *negative sign* means here — it is not that Jules is going backwards on her bike.

(e) How fast is Jules actually pedalling on N to P? Explain why that number is positive even though the rate of change is negative.

(f) Which segment is steepest, and what does that tell you about the ride?

6. A survey drone rises from the ground, hovers, climbs again, hovers again, then descends. The graph shows its altitude against time.



(a) Find the rate of change of altitude over the first 20 s, with units.

(b) Find it again for the climb from 50 s to 70 s. Which climb is faster, and how does the graph show it without any calculation?

(c) Find the rate of change on the descent, with units, and say in words what the negative sign tells you.

(d) For how many seconds in total is the altitude constant?

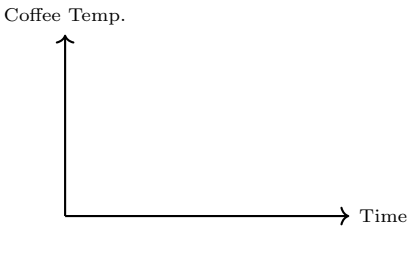
(e) The descent is steeper than either climb. Write one sentence comparing how fast the drone comes down with how fast it went up.

(f) While the graph is flat the drone's *altitude* is not changing. Does that prove the drone is not moving at all? Explain.

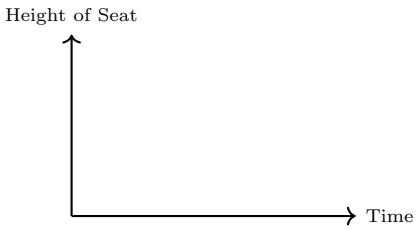
Part C — Sketching a Graph From a Story

7. Sketch a graph on each set of axes. There is no scale, so only the **shape** matters: where it rises, falls, stays flat, bends or repeats. Under each sketch write one sentence saying what makes your shape the right one.

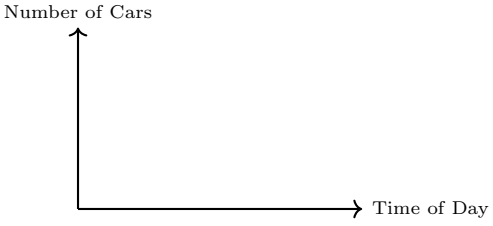
(a) A mug of hot coffee is left on a desk in a warm room and forgotten all afternoon.



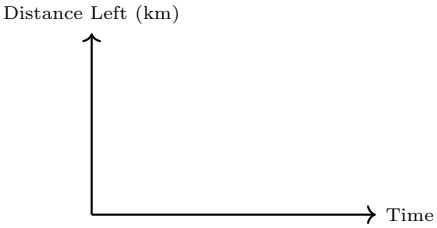
(b) One seat on a Ferris wheel turning at a steady rate, for three complete turns, starting from the loading platform at the bottom.



(c) The number of cars in a school parking lot from 7:00 a.m. to 5:00 p.m. on a school day.

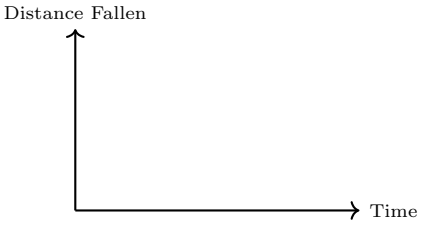


(d) A runner jogs a straight 5 km course at one steady pace. Graph the distance she still has left to run.

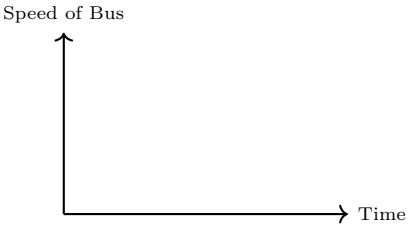


8. Sketch these. Each one has a detail that is easy to get wrong — the sentence you write underneath should say what it is.

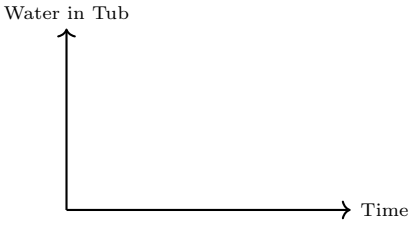
(a) A stone is dropped from a bridge and falls until it hits the water. Graph the distance it has fallen. (Careful: a falling stone does not fall at a steady speed.)



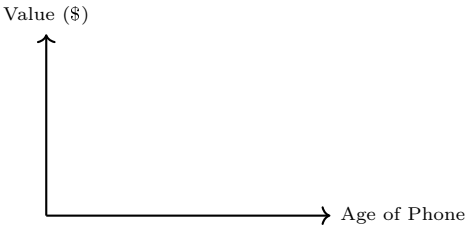
(b) A bus runs a route with three stops in it. Graph the bus's *speed* from the moment it leaves the depot until it reaches the third stop.



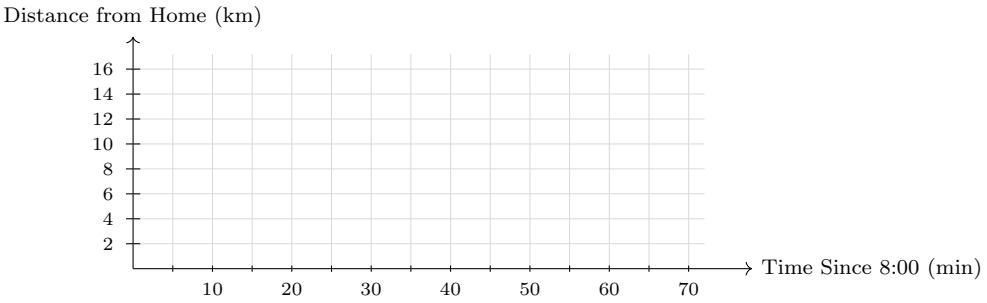
(c) A tap fills a bathtub at a steady rate. The tap is turned off, someone bathes for a while, and then the plug is pulled and the tub empties faster than it filled.



(d) A phone loses value quickly in its first year and more slowly after that, but a working phone is never quite worthless. Graph its value against its age.



9. This sketch has a scale, so the numbers matter. Sanjay leaves home at 8:00 a.m.
He walks 2 *km to the bus stop, arriving at 8:20. He waits* 10 *minutes. The bus then carries him* 12 *km further from home, arriving at 8:50. From there he walks the last* 1 *km to work, arriving at 9:05.*



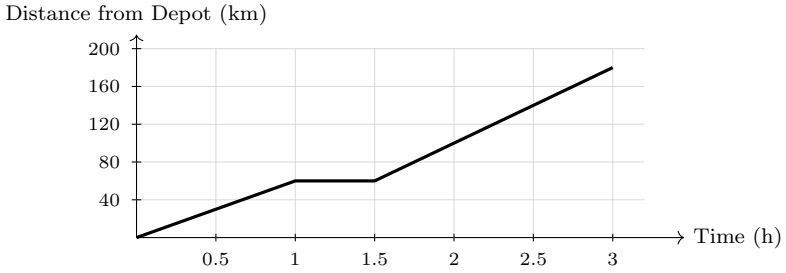
(a) Draw the graph of Sanjay's distance from home on the grid above.

(b) Find the rate of change of each of the four parts, in km/h.

(c) Which part of the journey has the steepest segment, and why would you expect that one to be steepest?

Part D — The Same Journey on Two Different Graphs

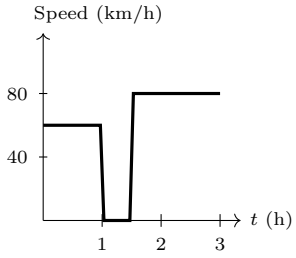
10. A delivery van leaves the depot, stops to unload, then drives on. Its **distance from the depot** is graphed below.



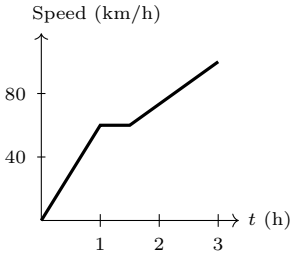
(a) Find the van's speed on each of the three parts of the journey, in km/h.

(b) Which of the four **speed**-time graphs below shows the same journey? Give the reason each of the other three is wrong. (The near-vertical pieces are drawn steep rather than truly vertical on purpose — see question 14.)

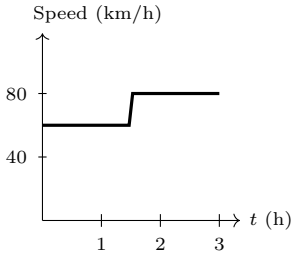
A.



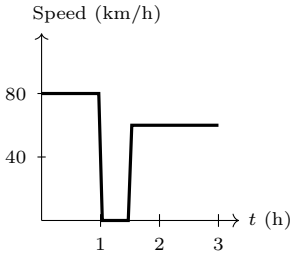
B.



C.



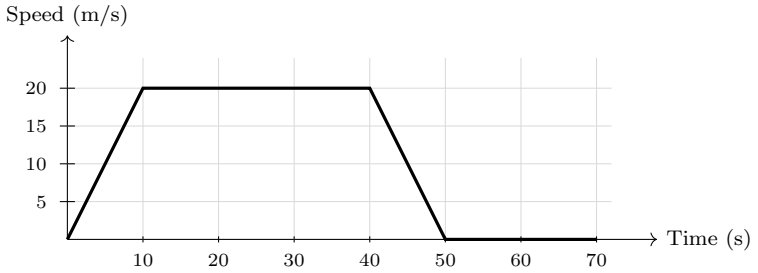
D.



11. This is the heart of the lesson. Complete the table: say what each feature means on a distance-time graph, and what the *same* feature means on a speed-time graph.

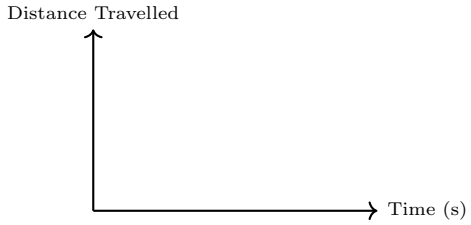
Feature of the graph	**On a distance-time graph it means. . .**	**On a speed-time graph it means. . .**
A horizontal segment above the time axis		
A segment lying *on* the time axis (value 0)		
A straight segment sloping up		
A straight segment sloping down		
A curve that gets steeper and steeper		

12. The graph below shows the **speed** of a light-rail train between two stations.



(a) Describe the journey in words, second by second. Name what is happening on each of the four parts.

(b) Sketch the matching **distance**-time graph for the same 70 seconds. It has no numbers on it — shape only.



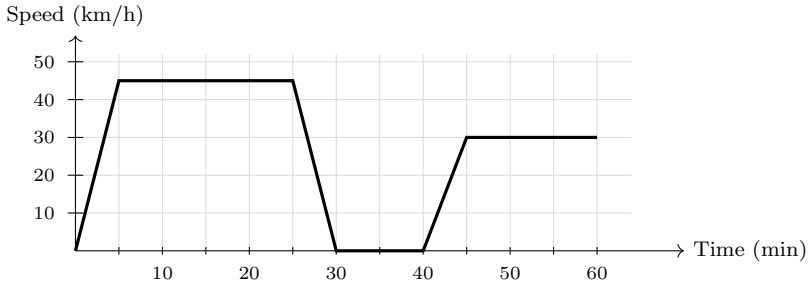
(c) The speed graph has two flat parts: one from 10 s to 40 s and one from 50 s to 70 s. They mean completely different things. Say what each one means.

(d) Your distance-time graph should also be flat somewhere. Over which seconds, and why is it flat only there?

(e) How far does the train travel between 10 s and 40 s? Explain how you got it from a graph of speed.

Part E — Find the Mistake

13. *(Error analysis)* The graph below shows the **speed** of a car on a trip across town. A student writes: *"Between* 5 *and* 25 *minutes the graph is flat, so the car wasn't moving during those* 20 *minutes."*



(a) Which graph was the student picturing when they said "flat means not moving"? State the rule they used and the rule they should have used.

(b) What is actually happening between 5 and 25 minutes? Include the speed.

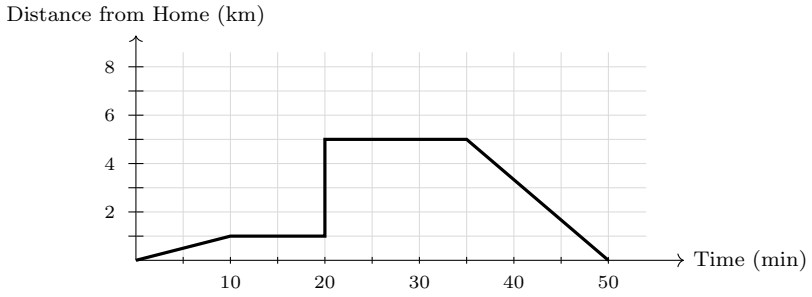
(c) There *is* a stretch of this trip where the car is not moving. Find it, and say what makes it different from the flat part the student chose.

(d) How far does the car travel between 5 and 25 minutes?

(e) Describe what the car is doing from 25 to 30 minutes, and from 40 to 45 minutes.

(f) Write a one-sentence rule the student could memorize so that this never happens again.

14. *(Error analysis)* Asked to graph her distance from home during an afternoon out, a student draws the graph below. It contains a vertical jump at 20 minutes.

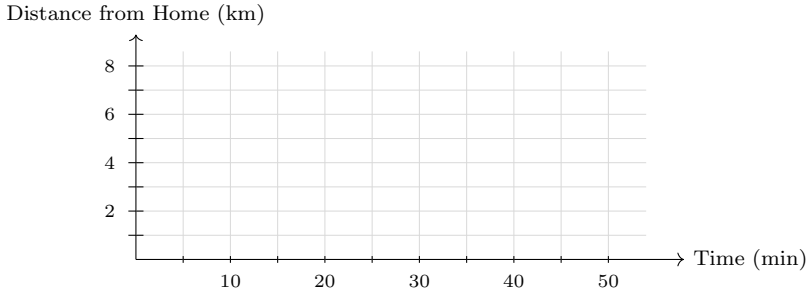


(a) Name the test this graph fails, and show exactly where it fails it.

(b) Put the failure into words about the outing itself: what would the student have had to do at 20 minutes?

(c) What rate of change would that vertical segment have? Explain why the arithmetic breaks down.

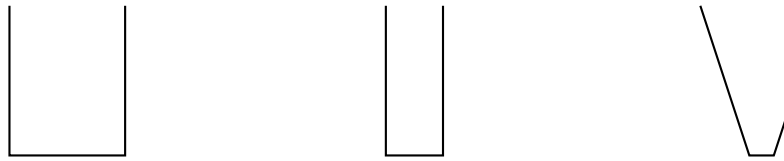
(d) The student meant that she covered the extra 4 km by bus, and that the bus ride took 5 minutes. Redraw the corrected graph on the grid below, keeping everything else the same. Then state the rate of change of each of the five segments, in km/h.



Part F — A Stretch: the Shape of the Container

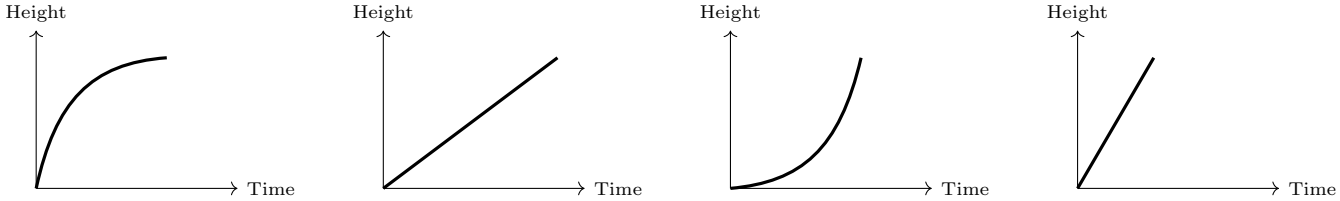
15. Water runs from the same tap, at the same constant rate, into each of the four containers below. All four are the same height and each is filled right to the top. The graphs show the **height of the water** in a container against time.

W. wide cylinder **X.** narrow cylinder **Y.** cone, point down **Z.** cone, point up



The four graphs are below, in a different order.

I. **II.** **III.** **IV.**



(a) Match each container with its graph, and give a reason for each that mentions the width of the container at the water line.

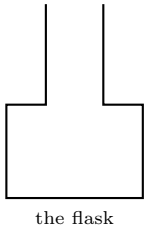
(b) Two of the graphs are straight lines. What do those two containers have in common, and why does that make the line straight?

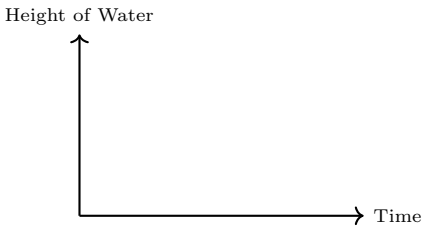
(c) Complete the rule: "the wider the container at the water line, the _____ the height rises." Explain it in terms of the water spreading out.

(d) The two cones are the same cone, one turned upside down, so they hold the same amount. What does that tell you about the two graphs, and does it contradict the fact that the graphs have different shapes?

(e) One straight-line graph is steeper than the other and also stops sooner. Explain why those two facts have to go together.

16. A flask is a wide cylinder for its bottom half and a narrower cylinder for its top half, joined by a flat shoulder. Water runs in at a constant rate.
(a) Sketch the height of the water against time, from empty to full.





(b) Your graph should have a sharp corner, not a smooth bend. Say where it is and why the change is sudden.

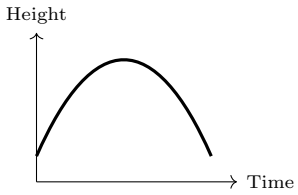
(c) Which part of your graph is steeper, and what feature of the flask makes it so?

(d) The two halves are the same *height*, but they do not take the same *time*. Which half takes longer to fill, and why?

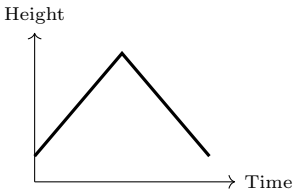
Part G — Two for the Test

17. *(Multiple Choice)* A ball is thrown straight up and caught again at the height it was released from. Which graph best shows the **height of the ball** against time?

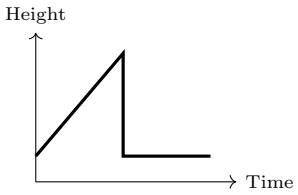
A.



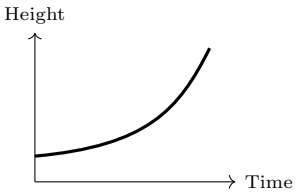
B.



C.



D.



18. *(Numerical Response)* On a distance-time graph, time is measured in minutes and distance in kilometres. A straight segment of the graph joins the point $(10, 4)$ to the point $(50, 18)$. The rate of change of that segment, in km/h, is _____. *(Record your answer as a three-digit number.)*

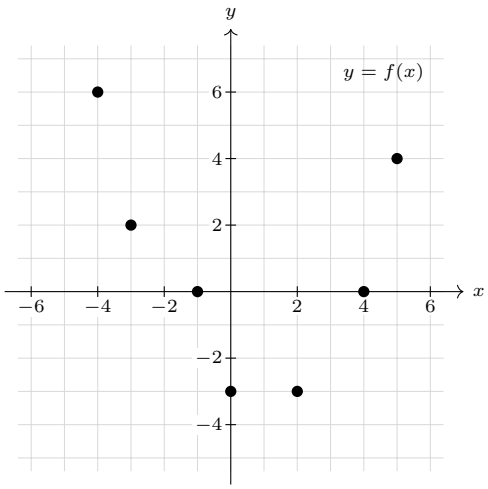
Lesson 9 — Extra Practice

Relations and Functions Unit Review

This is the unit's cumulative set, so nothing here stands on one lesson alone — every question asks you to carry a relation across two or more ideas. Keep the notation exact: a finite set of values is a ***list*** *in braces, each value written once and in increasing order, e.g.* $D = \{-2, 0, 3\}$*; an infinite set is* ***set-builder****, e.g.* $R = \{y \mid -1 \leq y < 5,\ y \in R\}$*, with* $\in I$ *in place of* $\in R$ *whenever the quantity can only be a whole number. A hollow endpoint means a strict inequality.* $f(3) = 7$ *and "*$(3, 7)$ *lies on the graph of* $y = f(x)$*" say the same thing.*

Part A — One Relation, Every Question

1. The graph below shows every point of a relation f. The dots are separate and are not joined. Answer every part from the graph.



(a) Write the relation as a set of ordered pairs.

(b) State the domain and the range. Two outputs are each produced twice — say how that shows up in your range.

(c) Is f a function? Name the test you used and say what you checked.

(d) State $f(-3)$ and $f(2)$.

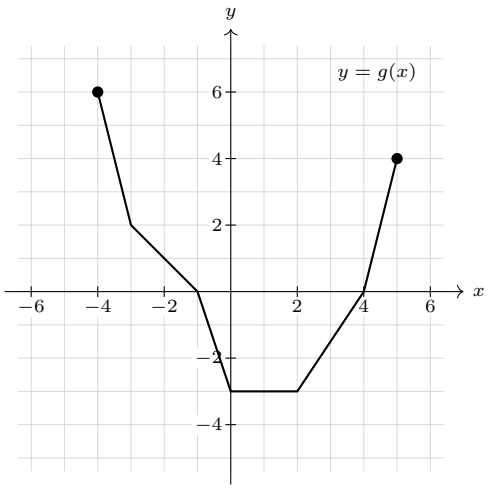
(e) Solve $f(x) = -3$, then solve $f(x) = 6$. Why does one have two answers and the other one?

(f) State the x-intercept(s) and the y-intercept as ordered pairs.

(g) Explain why "solve $f(x) = 0$" and "find the x-intercepts" are the same instruction, and why "evaluate $f(0)$" is a different one.

(h) Is this relation discrete or continuous? Say how the graph tells you, and name one number that is *not* in the domain even though it lies between two numbers that are.

2. The seven points of question 1 are now joined, in order from left to right, by straight segments. Everything about the relation has changed except the dots themselves.



(a) State the domain and the range of g. Both now need set-builder notation — explain why.

(b) The range of f in question 1 was a list of five numbers. Name a number that is in the range of g but not in the range of f, and say where on the graph it is produced.

(c) Solve $g(x) = 2$. One of the two answers is not a whole number — find it exactly, using the slope of the segment it lies on.

(d) Solve $g(x) = -3$. Compare your answer with $f(x) = -3$ from question 1(e) and explain the difference.

(e) Is g a function? State the intercepts of g, and say whether joining the dots changed them.

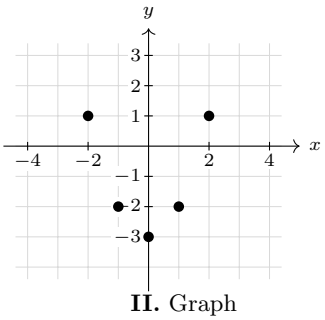
Part B — Domain, Range and Notation

3. One relation is shown below in four ways: a table, a graph, a mapping diagram, and an equation with a stated domain.

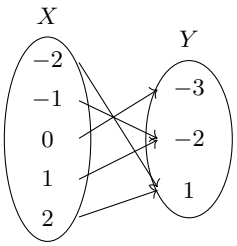
x	-2	-1	0	1	2
y	1	-2	-3	-2	1

I. Table of values

IV. Equation: $y = x^2 - 3$, with domain $\{-2, -1, 0, 1, 2\}$



II. Graph



III. Mapping diagram

(a) State the domain and range from the table, then from the graph. Do they agree?

(b) State the domain and range from the mapping diagram. The right-hand oval holds three numbers while the left holds five — explain why that is not an error.

(c) Use the equation to compute all five outputs and confirm they match the other three representations.

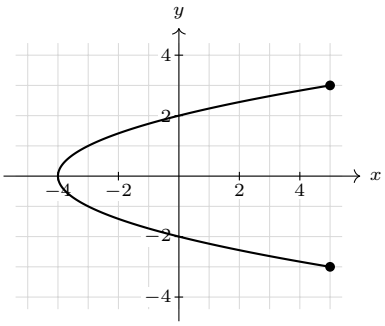
(d) Is the relation a function? Answer twice — once from the mapping diagram and once from the graph — naming the test in each case.

(e) Now delete the stated domain and treat $y = x^2 - 3$ as an equation on its own. State the new domain and the new range, in set-builder notation.

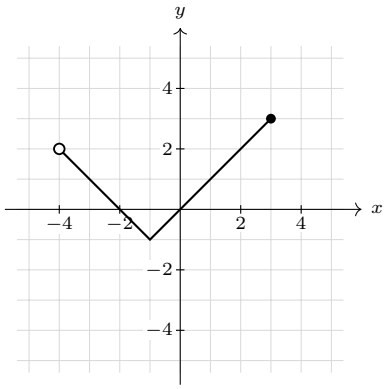
(f) Which of the four representations could *not* have told you the answer to (e), and why?

4. For each graph state (i) the domain, (ii) the range, (iii) whether the relation is a function, and (iv) every x-intercept and every y-intercept, as ordered pairs. If an intercept does not exist, say so.

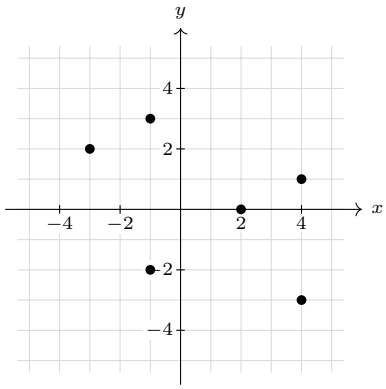
(a)



(b)



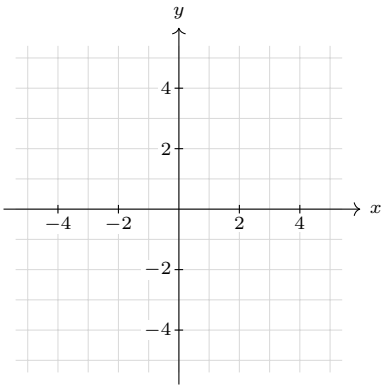
(c)



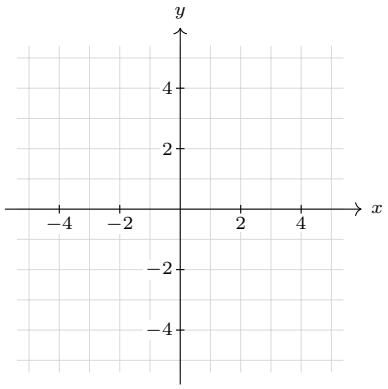
5. In part (a) of question 4 the relation has two y-intercepts. Explain why that fact alone settles the question of whether the relation is a function, without drawing a single vertical line.

6. Sketch a relation on each grid to meet the conditions given. Mark endpoints solid or hollow as required, and state anything the question asks you to state.

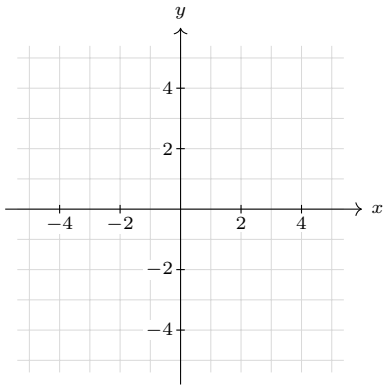
(a) A **function** with $D = \{x \mid -3 \le x \le 5,\ x \in R\}$, $R = \{y \mid -2 \le y \le 4,\ y \in R\}$, exactly two x-intercepts and one y-intercept.



(b) A relation that is **not** a function, with exactly the same domain and range as (a). Then name the one feature that makes the difference.



(c) A function with $D = \{x \mid x \in R\}$ and $R = \{4\}$. Then state both intercepts, or say which one does not exist.



Part C — Functions and Function Notation

7. Let $f(x) = 5 - 2x$ and $g(x) = x^2 - 3x$. Work exactly.

(a) Evaluate $f(-4)$ and $g(-3)$.

(b) Solve $f(x) = 13$. Then say what your answer to (a) already told you about this solution.

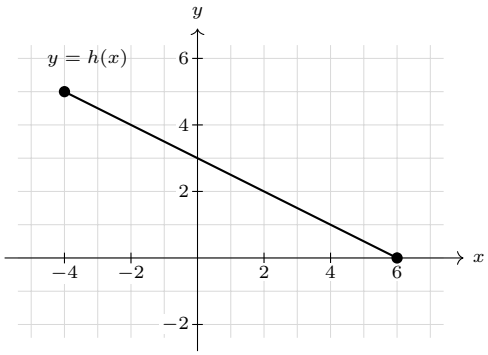
(c) Solve $g(x) = 4$. Explain why this one has two answers while (b) has one.

(d) Simplify $f(a + 2)$ and $g(2t)$.

(e) State the x-intercepts and the y-intercept of the graph of g, using function notation to say how you found each.

(f) State the domain and range of f, and the domain of g, with no restriction applied.

8. The graph below is the whole graph of a function h. Its equation is $h(x) = -\frac{1}{2}x + 3$.



(a) State the domain and range of h. Both bounds of the range come from the two endpoints — say which endpoint gives which.

(b) Read $h(0)$ and $h(4)$ off the graph, then check both by substituting into the equation.

(c) State both intercepts as ordered pairs. One of them is also an endpoint — which, and does that change the answer?

(d) Solve $h(x) = 1$ from the graph and again from the equation.

(e) A line usually has range $\{y \mid y \in R\}$. Explain in one sentence why this one does not.

9. Each function below is linear, but you are not given its equation.

(a) f satisfies $f(2) = 1$ and $f(5) = 10$. Determine $f(x)$, then state $f(0)$, the y-intercept and the x-intercept of the graph of f, and the domain and range of f.

(b) The graph of h has x-intercept $(-4, 0)$, and $h(2) = 9$. Determine $h(x)$, then solve $h(x) = 15$ and state the y-intercept.

Part D — Graphs in Context

10. A tablet plays video until its battery is flat. The charge remaining, C percent, after t hours of playback is modelled by $C(t) = 100 - 12.5t$.

(a) Determine $C(3)$ and say what it means.

(b) State the C-intercept and the t-intercept of the model, and interpret each in words.

(c) Solve $C(t) = 30$. Write the question this answers in plain English.

(d) State the practical domain and the practical range, in set-builder notation.

(e) A classmate says the domain is $\{t \mid t \in R\}$ "because the equation works for any t." Give a reason the left bound is wrong and a different reason the right bound is wrong.

(f) The number of *videos* watched is a different variable. Say whether it is discrete or continuous, and which of $\in R$ or $\in I$ its domain would use.

11. A school prints yearbooks. The profit, P dollars, on n yearbooks sold is $P(n) = 32n - 4800$, and the print run is 400 books.

(a) State $P(0)$ and explain what a negative value means here.

(b) Solve $P(n) = 0$ and give the usual name for that value of n.

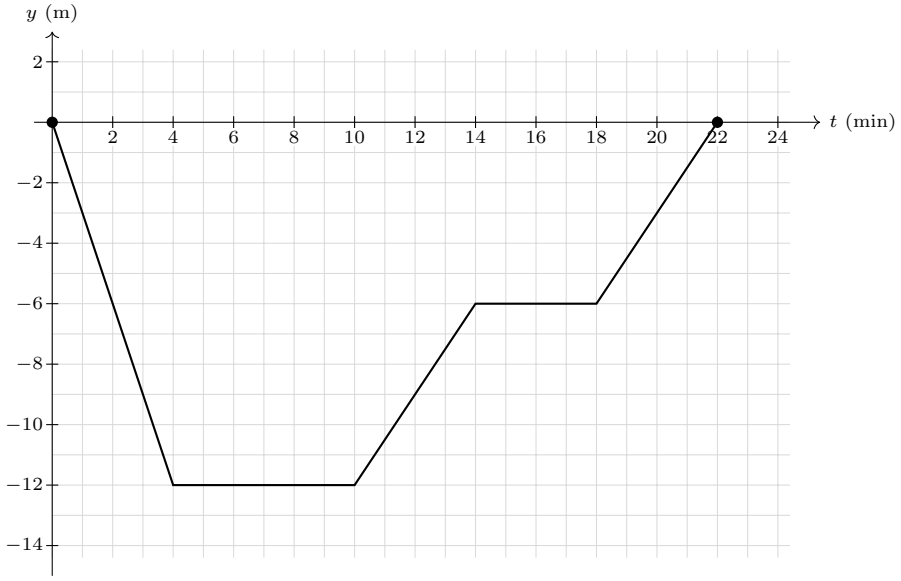
(c) Determine $P(220)$, and solve $P(n) = 3200$.

(d) State the practical domain. Say why it uses $\in I$ and not $\in R$.

(e) A student writes $R = \{P \mid -4800 \le P \le 8000,\ P \in R\}$. Both bounds are correct. Explain what is wrong, and give one value strictly between the bounds that the model never produces.

(f) Which of questions 10 and 11 has a graph made of separate dots, and which is an unbroken line? Justify from the variables, not the equations.

12. A diver's position relative to the surface of a lake, y metres, is graphed against time t minutes. Below the surface counts as negative.



(a) Writing the relation as $y = d(t)$, state $d(2)$, $d(6)$ and $d(16)$.

(b) State the domain and the range, in set-builder notation, with the correct variable letters.

(c) Is d a function? Give the test and what it shows on this graph.

(d) Solve $d(t) = -9$.

(e) Solve $d(t) = -6$. This answer is not a short list — describe it carefully and say which part of the graph forces that.

(f) State the t-intercepts and the y-intercept, and say what each one means to the diver.

(g) Over which interval(s) is d increasing fastest? Give the rate of change with units, and explain what a positive rate means for a diver.

(h) The deepest point is reached over a whole interval, not at a single instant. State that interval, and say why the range still has just one lower bound.

Part E — Reasoning and Error Analysis

13. Decide whether each statement is true or false. If it is false, give a specific counterexample — a relation, a graph or a pair of numbers, not just a sentence.

(a) Every relation is a function.

(b) A vertical line is a function.

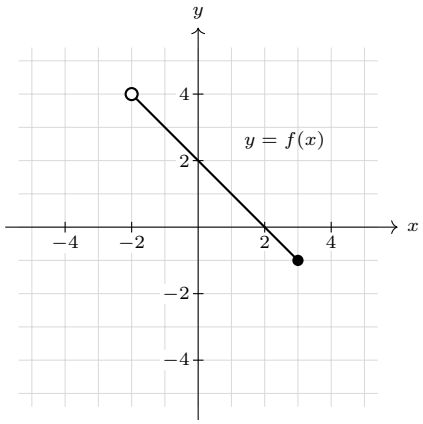
(c) The domain of a real-world model is always $x \geq 0$.

(d) A function has at most one y-intercept.

(e) A function has at most one x-intercept.

(f) If a relation is discrete, its domain must be written with $\in I$.

14. A student is given the graph below and writes the four answers shown. Every one of them is wrong. Identify the error in each and write the corrected version.



(a) "$D = \{y \mid -2 \leq y \leq 3,\ y \in R\}$."

(b) "$R = \{y \mid -1 \leq y \leq 4,\ y \in R\}$."

(c) "$f(0) = 2$, so the x-intercept is $(0, 2)$."

(d) "This is not a function, because the left endpoint is hollow."

15. Three more claims about function notation. Each is false. Say why, and back each one up using $f(x) = 3x - 1$.

(a) "$f(x)$ means f multiplied by x."

(b) "$f(2) + f(3) = f(5)$."

(c) "To solve $f(x) = 7$ you substitute 7 for x."

Part F — Challenge

16. A line has x-intercept $(6, 0)$ and y-intercept $(0, -4)$.

(a) Determine its equation in the form $y = mx + b$, and rewrite it in function notation as $k(x)$.

(b) The line is now restricted so that its range is $\{y \mid -4 \leq y \leq 2,\ y \in R\}$. Determine the restricted domain exactly.

(c) With that restriction in place, does the graph still have both intercepts? Justify.

17. A function f has a parabola for its graph. The parabola opens upward, its x-intercepts are $(-1, 0)$ and $(5, 0)$, and the smallest value f ever takes is -9.

(a) Determine $f(x)$ in the form $ax^2 + bx + c$. Show how the two intercepts and the minimum each pinned down part of the answer.

(b) State the y-intercept, the domain and the range.

(c) Solve $f(x) = -9$ and $f(x) = 7$. Explain why one of these has a single solution and the other has two.

18. *(Multiple Choice)* The whole graph of a function f is the straight segment from $(-5, -2)$ to $(3, 6)$, hollow at $(-5, -2)$ and solid at $(3, 6)$. Exactly one statement below is **false**.

A. $D = \{x \mid -5 < x \leq 3,\ x \in R\}$
B. $f(0) = 3$
C. The x-intercept is $(-3, 0)$
D. $R = \{y \mid -2 \leq y \leq 6,\ y \in R\}$

19. *(Multiple Choice)* The function h is defined by $h(x) = 5 - 3x$ on the domain $\{-2, 0, 1, 4\}$. The range of h is

A. $\{y \mid -7 \leq y \leq 11,\ y \in R\}$
B. $\{-7, 2, 5, 11\}$
C. $\{11, 5, 2, -7\}$
D. $\{y \mid -7 \leq y \leq 11,\ y \in I\}$

20. *(Numerical Response)* The graph of $g(x) = -x^2 + 6x + 16$ has x-intercepts $(a, 0)$ and $(b, 0)$ with $a < b$, and y-intercept $(0, c)$. The greatest value g ever takes is d. The value of $b + c - a - d$ is _____. *(Record your answer in the numerical response box from left to right.)*